Carbon Taxation of Food: Consumer Responses and Distributional Effects

Ondine Berland*

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Abstract

Food consumption accounts for a quarter of household carbon footprints, comparable in magnitude to housing and transport. Food products vary widely in carbon intensity, creating scope to reduce emissions by changing the composition of food baskets. A carbon tax could encourage such changes, but its effectiveness and incidence depend on what households purchase before the tax and how they adjust their purchases in response to higher prices. This paper combines granular household purchase data with product-level emission intensities to study these questions. The distribution of carbon footprints exhibits no clear income gradient (correlation = −0.03), and high-emitting products, particularly red meat, are purchased by most households. Reduced-form estimates show that higher red-meat prices mainly reduce purchased quantities, with no detectable change in the probability of purchase. Estimates from a structural demand system indicate that red meat is highly price-elastic, and that households substitute toward less carbon-intensive categories when its price increases. Simulations show that a carbon tax of €44.6 per ton of CO_2e reduces food emissions by 7.9%. Uniform redistribution of tax revenues brings average net tax payments close to zero in every income decile.

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*London School of Economics and Political Science. Email: o.berland@lse.ac.uk

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1 Introduction

Reducing the greenhouse gas (GHG) emissions associated with what households eat is an important component of climate mitigation. Food consumption accounts for a substantial share of household carbon footprints, 19% in the US and 24% in France, a contribution comparable in magnitude to housing and transport (Barbier et al., 2019; Song et al., 2019). Food products vary widely in carbon intensity, creating scope to reduce emissions by changing the composition of food baskets. A carbon tax could foster such changes by raising the relative prices of more carbon-intensive foods. Assessing the potential of such a policy requires understanding how food carbon footprints vary across households, how households respond to price changes, and how the tax burden is distributed.

Food baskets differ substantially across the income distribution (Darmon and Drewnowski, 2008; Allcott et al., 2019b), yet little is known about how food carbon footprints are distributed across households. Higher incomes may increase demand for relatively expensive, carbon-intensive foods, such as red meat. Conversely, higher-income households may place greater weight on environmental attributes and select less carbon-intensive products within categories. Assessing how these differences translate into carbon footprints requires detailed data linking household food purchases to product-level emission intensities, which are rarely available together.

This paper studies the distribution of food carbon footprints and evaluates the emissions effects and distributional incidence of a food carbon tax in France. To do so, I link detailed household food purchases to product-level emission intensities. I find that food carbon footprints are weakly correlated with income (corr = -0.03), with high-emitting households found throughout the income distribution. Red-meat consumption is an important source of variation in emissions across households. I study how red-meat purchases respond to prices and estimate a demand system to capture substitution across food categories. Simulations suggest that a food carbon tax of €44.6 per ton of CO_2 equivalent (tCO_2e) could reduce food carbon footprints by 7.9%. However, simulated tax payments are similar in absolute terms across income deciles, making the tax regressive as a share of income. A uniform rebate per consumption unit brings average net payments close to zero in each income decile.

I construct a novel dataset for France by matching household food purchases from the Kantar World-Panel to emission intensities from the Agribalyse life-cycle assessment database. The Kantar data are transaction-level: households record the quantity, expenditure, and attributes of each food product they buy for home consumption, which accounts for close to 80% of household food expenditure in France. My sample covers over 20 million transactions by 10,006 households between 2016 and 2019, including purchases of unpackaged items. Agribalyse provides emission intensities per kilogram for standardized products, such as plain yogurt or minced beef, rather than for broad food categories. A matching procedure links household purchases to product-level emission intensities for products accounting for 98% of transactions. Linking household purchases to product-level emission intensities allows me to measure food carbon footprints from purchased quantities and the composition of food baskets rather than expendi-

ture. This distinction matters because higher food expenditure may reflect higher prices rather than more carbon-intensive consumption. I calculate annual household food carbon footprints by multiplying the quantity purchased of each product by its associated carbon emission intensity and summing across the entire food basket. Food carbon footprints are expressed per consumption unit, a measure that adjusts household size by assigning different weights to adults and children to account for household composition and economies of scale.

I first document the distribution and determinants of household food carbon footprints, which average 2.20 tCO_2e per consumption unit per year (SD=0.76). Differences across households arise along two margins: the total quantity of food purchased and the composition of the food basket—that is, how purchases are allocated across products with different carbon intensities. Red meat illustrates why this second margin matters: it accounts for only 1.3% of purchased volume and 5.6% of food expenditure, but 19.1% of food emissions. Almost all households consume meat (only 0.75% of household-quarters contain no meat purchases). High emitters therefore do not form a clear subgroup of the population that buys most of the carbon intensive foods. The correlation between income and food carbon footprints is small -0.03 ($p<0.01$), and the partial correlation conditional on household characteristics is 0.03 ($p<0.01$). A concern with scanner data is that panels may under-represent some households, in particular top earners, so the weak gradient could reflect sample composition rather than household food choices. Two pieces of evidence suggest otherwise. Applying expenditure-based emission factors to the same households, by contrast, produces a clear income gradient. In addition, I reproduce the analysis using a representative national food consumption survey (INCA3), which includes food consumed away from home, and find again a flat income gradient.

I next study how households adjust their food baskets when prices change. Reduced-form regressions for red meat, the most emission-intensive category, provide transparent evidence on responses along the extensive and intensive margins, without imposing the structure of a full demand system. They show no effect on the extensive margin: the probability of purchase is unchanged (estimate of 0.019, $p>0.1$). On the intensive margin, conditional on purchasing, a 1% increase in the price of red meat reduces expenditure by 0.16% and purchased quantities by 0.18%. Households therefore respond primarily by purchasing smaller quantities, rather than by exiting the category. These single-category estimates, however, are silent on what households buy instead. Yet the net emissions effect of a price change hinges on these substitutions. To capture them, I estimate a Quadratic Almost Ideal Demand System (QUAIDS) over 13 food categories. Red meat is the most price-elastic category, and cross-price elasticities point to substitution toward less emission-intensive products, notably other meats, dairy, and prepared foods. These interactions highlight the importance of accounting for changes in the composition of the entire food basket to study the effects of a food carbon tax.

I set the tax at €44.6/tCO_2e, the level of the carbon component of French excise duties on fossil fuels, and apply it to each product in proportion to its emission intensity. I simulate three scenarios: a benchmark in which the tax covers all food products, and two narrower taxes restricted to meat and to

red meat, respectively. In the benchmark, the tax reduces the aggregate carbon footprint of food by 7.9% and raises the food price index by 4.6%. Restricting the tax to meat products achieves a 5.0% reduction while raising the food price index by only 1.7%. Taxing red meat alone delivers more than half of the benchmark's emission reduction (4.3%) for about a fifth of its price increase (0.9%).

Under the benchmark scenario, annual tax payments are €92 per consumption unit in the bottom income decile and €88 in the top decile. However, these payments represent 1.1% and 0.2% of income, respectively, making the tax regressive. Uniform revenue recycling brings average net tax payments close to zero across income deciles, although it does not fully compensate households for the welfare cost of the price increase. Tax incidence is heterogeneous within income deciles, showing that income alone is insufficient to identify which households bear the cost of the tax. These alternative tax bases illustrate the resulting policy trade-off: targeting meat, and red meat in particular, preserves a substantial share of the reduction in emissions while limiting food-price increases and household tax payments.

Related literature. This paper makes several contributions. First, I complement the literature on environmental Engel curves, which studies the relationship between household income and the emissions embodied in the goods and services households consume (Levinson and O'Brien, 2019). This relationship is a key determinant of the incidence of carbon pricing. Most studies rely on input-output models that apply aggregate emission intensities per euro to household expenditure, and find that carbon footprints increase with income (Chancel, 2022; Pottier, 2022). This approach raises specific difficulties for food. Within a food category, prices vary with quality, brand, and availability (Handbury and Weinstein, 2015), whereas emissions mainly depend on the quantity purchased and on the type of food, for instance animal products versus plant-based substitutes (Poore and Nemecek, 2018). Expenditure-based measures may therefore attribute larger carbon footprints to households that pay higher prices, even when they purchase the same quantities of products with the same emission intensities (André et al., 2024). Moreover, these measures rely on a coarse classification of food products, which assigns the same emission intensity to products whose emissions differ widely (Clark et al., 2022). As a result, they cannot capture how differences in the composition of food baskets translate into differences in food carbon footprints. Instead, I measure food carbon footprints from purchased quantities, using transaction-level data matched to fine-grained product-level emission intensities. In doing so, I revisit within-country inequality in food carbon footprints. Comparing two measures for the same households, I show that quantity-based food carbon footprints are only weakly correlated with income, whereas an expenditure-based measure attributes a 35% larger footprint to the richest income decile than to the poorest. The weak income gradient is consistent with studies in nutrition and public health that also measure food carbon footprints from quantities (Reynolds et al., 2019; Perignon et al., 2023). I go further by linking these footprints to household price responses and to the incidence of a food carbon tax.

Second, I contribute to the literature on corrective taxation. Since the seminal work of Pigou (1920), a large literature has studied the taxation of environmental externalities, with carbon taxes on energy as

a main application. This literature emphasizes both behavioral responses and distributional incidence: carbon taxes are typically regressive because energy budget shares decline with income, while revenue recycling can mitigate vertical regressivity, and burdens may vary substantially within income groups (Metcalf, 2009; Douenne, 2020). Food raises distinct issues. Nearly all food products generate emissions, at very different intensities, and many are substitutes or complements in consumption, so the effect of a food carbon tax depends on how households reallocate expenditure across the entire food basket. Food carbon taxation also differs from corrective taxation of specific products such as alcohol and sugar-sweetened beverages. In these settings, externalities are often concentrated among heavy consumers, making the identification and targeting of these consumers central (Griffith et al., 2019; Dubois et al., 2020; Allcott et al., 2019a). In contrast, high-emission foods are widely purchased and high emitters are found throughout the population, leaving less scope to target a distinct group of consumers.

Despite growing interest in taxing food emissions (Funke et al., 2022), existing evidence remains incomplete. Demand estimation either restrict substitution to meat products (Bonnet et al., 2018; Roosen et al., 2022) or estimate responses at an aggregated level (Caillavet et al., 2016), while distributional analyses either abstract from behavioral responses (Klenert et al., 2023) or focus on aggregate emission reductions and mean welfare effects rather than how burdens vary across households (Plinke et al., 2026). I bring these two dimensions together by estimating household-level demand responses across the entire food basket and using them to study both the environmental effects and the vertical and horizontal incidence of a food carbon tax.

The remainder of this paper is structured as follows. Section 2 describes the data and key stylized facts. Section 4 presents the reduced-form results. Section 5 presents the structural demand model and estimation results. Section 6 discusses the counterfactual and associated results. A last section concludes.

2 Data

In this section, I describe the main data sources used in the paper. I outline the construction of an original database combining two datasets to study household food carbon footprints.

2.1 Food Purchases

Food purchase data are from the homescan dataset of *Kantar WorldPanel* for France from 2016 to 2019. For data availability reasons, this study is restricted to food-at-home purchases, which account for about 78% of food expenditure in France on average.[1] This dataset is well-suited for analyzing food carbon footprints due to its high level of granularity and inclusion of both expenditure and volumes. Detailed household characteristics, including household size, income, age, and employment status, allow me to

[1]Food-away-from-home represents on average 22% of total food expenditure (median 11.2%). Source: author's calculations based on the latest wave (2017) of the French household budget survey (*Budget de Famille*).

examine how these footprints vary across households.

Households participating in the panel are required to systematically record details of all purchased products after each shopping trip, resulting in transaction-level data that contains, for each purchase, the brand, product attributes describing the item, a product identifier, and volumes and expenditures.[2] A subset of these households also records items sold loose, without a manufacturer product code, referred to as 'random-weight' items. These consist primarily of fresh fruits and vegetables, bread, meat, and fish, and account for approximately one-third of food sales in France.[3] Around 20,000 households participate in the panel each year, of which approximately 12,000 report random-weight purchases. Kantar defines active households as those reporting purchases for at least ten months of the year.

Sample restrictions. To obtain as complete an account of household food purchases as possible, I restrict the sample to that are both active and reporting random-weight items. These criteria address distinct sources of undercoverage: the active-household criterion limits incomplete reporting across the year, while the random-weight criterion captures unpackaged products that would otherwise be omitted. The resulting sample contains 10,006 households observed for an average of 2.4 years (ranging from one to four years).[4] These restrictions bring the sample closer to the French population than the full Kantar. Appendix Table A.1 compares socio-demographic moments in the full panel and in the selected sample with the corresponding figures for French households, drawn from the Population Census, the Labor Force Survey, and the Tax and Social Income Survey. Relative to the full panel, the selected sample is older (52.3 against 47.8 years), smaller in household size (2.19 against 2.72 persons) and less college-educated (41.0% against 45.6%). On each of these dimensions it moves toward the national figures of 53.0 years, 2.20 persons and 40.0%. Mean income per consumption unit remains 7–8% below the national figure.[5] This difference in income levels is unlikely to drive the main results. The demand estimates are identified from price and expenditure variation within the sample. For the relationship between income and food carbon footprints, Section 3.3 compares the results with those obtained from INCA3, a survey representative of the French adult population that also covers food consumed away from home, and finds a similar pattern.

Table A.2 compares food-at-home expenditure in the analysis sample with that reported in the 2017 wave of the French household budget survey (*Budget de Famille*). Expenditure per consumption unit is 8.9% lower in the Kantar, consistent with the tendency toward under-reporting in homescan data. Despite this difference in levels, basket composition is very similar: budget shares differ by no more than one percentage point across the main food categories. This similarity suggests that the Kantar accurately captures the relative allocation of food expenditure across the main categories.

[2]Barcodes are not provided; products are identified using Kantar product identifiers.

[3]Data on non-standardized products are from the French National Institute for Statistics and Economic Studies: https://www.insee.fr/fr/statistiques/4473482.

[4]Kantar provides corrections for outliers and implausible observations, such as unusually large purchase quantities, as well as period and household weights. Both are applied throughout the data processing.

[5]Part of this gap is definitional, as income in the Kantar is self-reported in brackets while the national figure is disposable income measured from tax and social records, and part reflects the documented undercoverage of top earners in homescan panels (e.g., Allais et al., 2010; Caillavet et al., 2019).

2.2 Environmental Impact of Food Products

Environmental impact data for food products are sourced from the Agribalyse database, version 3.0, produced by the French Ministry of Environment. It provides life-cycle assessment estimates for 2,480 standardized food products that are representative of the French market, corresponding to 1,150 distinct emission intensities; products sharing an identical value are typically varietal distinctions with no separate life-cycle assessment, such as blonde, coral, and green dry lentils. This level of detail exceeds that available in previous studies—around 20 categories in input-output data (Stadler et al., 2018), and 311 products in Caillavet et al. (2016) and Bonnet et al. (2018).

Agribalyse contains several environmental impact measures, including effects on climate change, water quality, air quality, and soil health, all expressed per kilogram of food product. I use Agribalyse's climate-change indicator, expressed in kilograms of CO_2 equivalent per kilogram of product ($\mathrm{kg\,CO_2e/kg}$).[6]

One limitation of Agribalyse is that it provides emission factors for standardized products rather than individual Kantar products. For example, a standardized product such as plain yogurt reflects the overall French market offer rather than the specific sourcing, recipe, or production methods of an individual yogurt. Each standardized product is assigned a single emission factor, calculated as a production-weighted average across the practices observed in the French market. This structure supports comparisons between standardized products but not within them: Kantar products matched to the same standardized product receive identical emission factors even when their actual production methods differ.

This limitation nevertheless leaves substantial variation in food carbon intensities observable in the data. Agribalyse covers a large range of standardized products representative of the French food market, and differences in emission intensity remain large even at high levels of aggregation. A variance decomposition of product-level emission intensities attributes 57% of the observed variation to differences across the 13 broad food categories used in the analysis and 43% to variation within them (Appendix Table B.2). Among meat products, 72% of the variation lies between red, white, and other meat rather than within these categories. Red meat, for example, has an average emission intensity of around 41 $\mathrm{kg\,CO_2e/kg}$, roughly four times that of white meat, while emission intensities within the red-meat category are comparatively concentrated. Some lower-emission categories display substantial within-category dispersion, such as 'Other Beverages', which include both water and coffee, but these categories contribute to a much smaller share of household food carbon footprints. Agribalyse intensities are in line with the global meta-analysis of Poore and Nemecek (2018), which covers about 38,700 farms in 119 countries. White meat averages 8.94 kg CO_2e/kg in my data, close to their global means for poultry (9.9) and pork (10.49). Red meat averages 41 kg CO_2e/kg, within their range for ruminant meat, which runs from 33 for beef from dairy herds to 99 for beef from beef herds. Clark et al. (2022) find a similar ordering for supermarket products in the United Kingdom and Ireland, with low impacts for sugary drinks, fruit, and bread and high impacts for meat, fish, and cheese.

[6] The CO_2 equivalent allows greenhouse gases to be compared based on their global warming potential.

I further examine whether some observable sources of heterogeneity within standardized products are likely to generate substantial measurement error. Product origin is particularly relevant for beef because of its high emission intensity and large contribution to household food carbon footprints. Beef purchased for home consumption in France is predominantly domestic: imports account for only 11% of beef sold in supermarkets and 15% of beef sold in butchers' shops, compared with 55% in out-of-home catering (GEB-Institut de l'Élevage, 2019). The first two figures are the relevant ones for this study, which covers food-at-home purchases. Imported beef also comes predominantly from neighboring European countries with broadly comparable production systems, while origins associated with markedly different systems account for only a small share of imports.[7] Import exposure is higher for poultry and ovine meat, but this category has a much lower emission intensity than beef, while ovine meat represents only a small share of total meat consumption (2.8% in 2019).[8] Origin-related differences are therefore unlikely to have a large effect on aggregate household food carbon footprints. Bellassen et al. (2022) find no significant difference in carbon footprints per ton between certified and conventional products overall, with the exception of organic plant-based products. Organic purchases represent only 2.4% of total food volume in the data and are concentrated in relatively low-emission categories such as fresh fruits and vegetables. Differences associated with certification are therefore also unlikely to generate large systematic errors in measured household food carbon footprints.

For these reasons, unobserved variation among Kantar products matched to the same standardized product enters household food carbon footprints as measurement error, but this error is not expected to be systematically correlated with household characteristics. I next describe the matching procedure between the Kantar and Agribalyse datasets.

2.3 Carbon Footprint of Food Purchases

Matching and random-forest predictions. To calculate food carbon footprints, I match each Kantar product to a standardized product in Agribalyse and assign the corresponding emission factor. Kantar records approximately 200,000 distinct product identifiers annually and provides detailed information on their brands and attributes. I first use a product-level matching table developed by NutriXConso, which links earlier Kantar product identifiers to the standardized food-product identifiers used by Agribalyse (De Mouzon and Orozco, 2011), and supplement it with manual matches. Because Kantar identifiers change over time, many products observed in 2016–2019 remain unmatched. I therefore train a random-

[7] Beef import origin is computed from the World Bank's World Integrated Trade Solution (WITS), which distributes UN Comtrade merchandise trade data on imports by partner country (HS codes 0201, fresh or chilled bovine meat, and 0202, frozen bovine meat). Figures are averaged over 2016–2019 to match the sample period. Imports originate overwhelmingly from neighboring European countries (the Netherlands, Ireland, the United Kingdom, Germany, and Poland), whose production systems are comparable to French ones, while countries with markedly more extensive systems such as Brazil account for only about 0.3% of the volumes of beef imports.

[8] Author's calculation based on official French Ministry of Agriculture statistics published by Agreste (2020), Table 1. https://agreste.agriculture.gouv.fr/agreste-web/download/publication/publie/SynCsm20359/consyn359202006ConsoViande.pdf.

forest classifier on 105,831 products with existing or manual matches in 2011. Using Kantar product attributes as predictors, the classifier assigns each of 569,295 previously unmatched product-year records to one of the thousand standardized products represented in the training data.

Manual matches also draw on standardized products absent from the 2011 training data, so the matched products correspond to between 1,047 and 1,062 distinct standardized products each year, more than the 939 the classifier can predict. Together, the pre-existing matching table, manual matches, and random-forest predictions cover Kantar products accounting for 98% of transactions over the study period. Appendix A.2 describes the procedure and validation results in detail. Because matching relies on product attributes rather than barcodes, it covers both packaged products and random-weight items.

I evaluate the classifier on 2016–2019 products whose correct match is already known from the NutriXConso table or manual matching: I hide the known match, predict it with the classifier trained on 2011 data, and compare the two. Appendix Table A.3 reports the results. Accuracy, the share of products assigned to the correct standardized product, is high (ranging from 92.3% to 93.5% across years). Accuracy is very high where errors would matter most for emissions, reaching 97.4% for beef. It is lowest for products such as seasonings. Because accuracy counts each Kantar product equally, it mainly reflects performance on standardized products matched to many Kantar products. I report another measure of the algorithm performance: the macro-F1, which measures performance separately for each standardized product and averages across them, so that rare standardized products count as much as common ones. The performance of the algorithm remains high (0.820 to 0.839).[9]

Household food carbon footprints. For each year, I compute a household's total food carbon footprint by summing, across all products purchased during that year, the quantity purchased multiplied by the product-specific emission intensity assigned through the matching procedure.[10] To make annual food carbon footprints comparable across households of different sizes and demographic compositions, I divide the household total by the number of adult equivalents (consumption units), calculated using the modified OECD equivalence scale. This scale assigns a weight of 1 to the first adult, 0.5 to each additional household member aged 14 or older, and 0.3 to each child under 14. Food carbon footprints are therefore expressed in kilograms of CO_2 equivalent (kg CO_2e) per consumption unit per year. Throughout the main analysis, "food carbon footprint" refers to this measure rather than the unadjusted household total.

[9] The F1 score is lower than accuracy because rare standardized products have few training observations and are therefore harder to predict.

[10] Quantities recorded in units other than kilograms, such as litres, are converted to kilograms before emission factors are applied.

3 Stylized Facts on Household Food Carbon Footprints

This section documents key features of household food carbon footprints. First, total food-related carbon emissions vary relatively little across households, whereas emissions from red meat exhibit greater variation. Second, food emissions are strongly shaped by basket composition: meat, and red meat in particular, contributes disproportionately to emissions relative to its share of expenditure or purchased volume. Third, food carbon footprints display no economically meaningful income gradient. Finally, higher-emitting households purchase larger quantities and tend to be older, more rural, and less educated, while stated environmental preferences are only weakly associated with observed emissions.

3.1 Distribution of Food Carbon Footprints

For each household, I calculate annual food carbon footprints by summing product-level emissions and dividing by the number of adult equivalents (consumption units), to account for economies of scale in consumption. Figure 1 shows the cumulative distribution of household food carbon footprints. The mean annual food carbon footprint is 2.20 tCO_2e per consumption unit, and is very close to the median of 2.09 tCO_2e.[11]

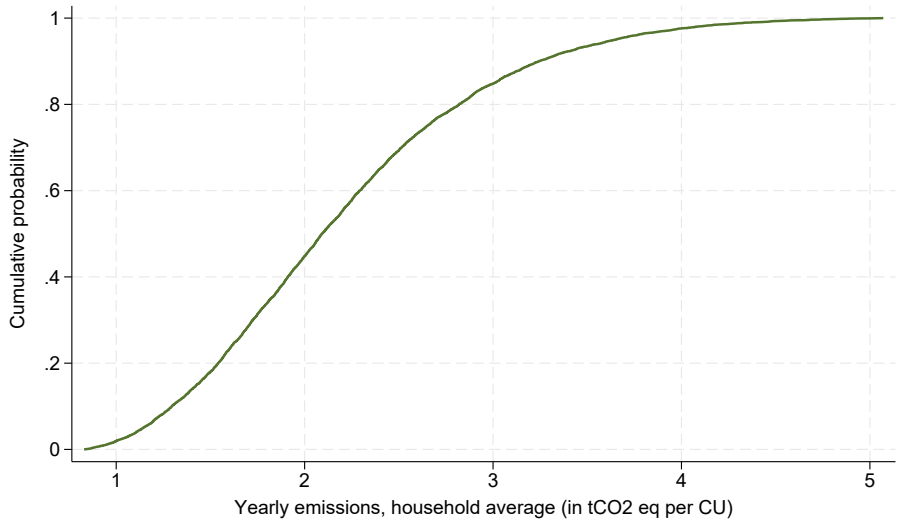


Figure 1: Cumulative Distribution of household food Carbon Footprints.

Notes: Average yearly emissions per consumption unit. The x-axis shows the range of yearly emissions per household, while the y-axis indicates the cumulative probability. The median is 2.09 yearly tCO_2e per consumption unit. Emissions are trimmed at the bottom and top 1 percentiles of the distribution. Own calculations based on Kantar 2016-2019 and Agribalyse data.

[11] Expressed per capita, the mean is 1.62 tCO_2e. This is below the national estimate of approximately 2.0 tCO_2e per capita implied by food accounting for 24% of an 8.3 tCO_2e household carbon footprint (Barbier et al., 2019). The difference reflects the scope of the scanner data, which cover food purchased for consumption at home. Barbier et al. (2019) estimate that approximately one meal in seven in France is consumed away from home.

The distribution is concentrated around its mean with low right skew. Its standard deviation is 0.76 tCO_2e per consumption unit, its skewness is 0.73, and its excess kurtosis is 0.35.[12] The distribution is close to that of a normal distribution and well below the values typically observed for income (McDonald et al., 2013), which usually exhibits strong skewness with fat tails. Unlike income or financial assets, food consumption is naturally bounded by how much households can consume. Food emissions consequently display fewer extreme values.

Households above the eighth decile emit more than 2.8 tCO_2e per consumption unit, and the top quintile accounts for 38% of aggregate food emissions. Appendix B.1 presents the Lorenz curve for household yearly food carbon footprints; the Gini coefficient is 0.19.

Red-meat emissions are more heterogenous across households. The Gini coefficient is 0.40, and the top quintile of red-meat emitters accounts for approximately half of aggregate red-meat emissions (Appendix Figure B.1). This greater concentration reflects differences in red-meat purchases within otherwise broadly diversified baskets, including whether red meat is purchased during a given period and how much is purchased.

3.2 Composition of Food Baskets

A household's food carbon footprint depends on the quantities purchased and the emission intensity of each product. The contribution of a category to total food carbon footprints may therefore differ from its share of food expenditure or purchased volume. Figure 2 presents the average emission intensity of the 13 product categories used in the analysis. These categories broadly correspond to supermarket aisles and are defined in Appendix Section A.3. The bars report average emission intensity per kilogram, weighted by purchased volumes, while the vertical lines indicate the 10th and 90th percentiles of the product-level distribution within each category. As shown in Section 2.2, most variation in emission intensity is across rather than within categories.

Red meat is by far the most emission-intensive category, with an average intensity of approximately 41 kg CO_2e/kg. This is about four times the intensity of the next-highest category. More generally, animal products rank among the most emission-intensive foods, whereas fresh fruit and vegetables, starches, and soft drinks have lower average intensities.

Meat contributes disproportionately to food carbon footprints: its share of emissions is larger than its shares of food expenditure and consumption volume. Table 1 reports the distribution of aggregate expenditure, purchased volumes, and emissions across the 13 categories. Red meat represents only 1.3% of purchased volumes and 5.6% of food expenditure, but accounts for 19.1% of food emissions. Taken together, red, white, and other meat account for 6.4% of purchased volumes and 22.8% of expenditure,

[12] The excess kurtosis is defined as the kurtosis minus three, which is the value that characterizes the normal distribution. Here the kurtosis is 3.35.

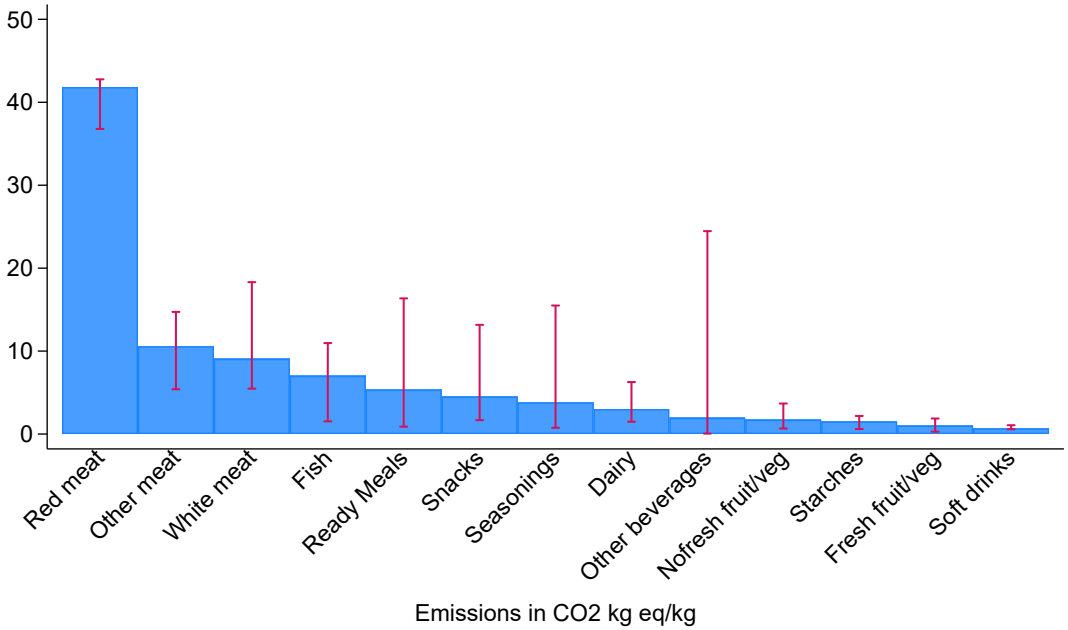


Figure 2: Emission Intensity per Food Product Category.

Notes: The large blue bars represent the volume-weighted averages, while the fine red bars display the 10th and 90th percentiles for each product (not volume-weighted to highlight the spread of emissions within each category). Appendix A.3 defines each food category. Own calculations based on Kantar 2016-2019 and Agribalyse data.

but 36.6% of food emissions. Including fish and dairy, animal products represent only 17.1% of purchased volumes but 49.2% of food emissions.

High-emission foods are not consumed by a small and separate group of households. At the quarterly level, 73% of household-period observations contain positive purchases in all 13 categories. Only 0.6% contain no meat purchases but some fish purchases, while very few (0.15%) show neither meat nor fish purchases.[13] Variation in food carbon footprints therefore arises within baskets that include nearly all food categories, through differences in the quantities purchased and the allocation of purchases across meat categories. These results highlight basket composition as a potential margin for reducing household food carbon footprints. Most households purchase from nearly all food categories, but emission intensities differ across categories, making the relative quantities purchased important for emissions. This motivates the study of a carbon tax, which, by changing relative food prices, can shift purchases towards less emission-intensive foods.

[13]These classifications are inferred from household purchases within a quarter and should not be interpreted as self-reported dietary identities. They are also measured at the household rather than individual level. For comparison, vegetarians represented 0.8% of the French population in 2020. Source: FranceAgriMer Report 2020, https://www.franceagrimer.fr/chiffre-et-analyses-economiques/vegetariens-et-flexitariens-en-france-en-2020.

Table 1: Shares of expenditure, purchased volume, and emissions by food category

Food category	Expenditure share (%)	Volume share (%)	Emissions share (%)
Red meat	5.60	1.33	19.12
Other meat	9.79	2.58	9.54
White meat	7.36	2.49	7.89
Fish	6.71	1.63	4.04
Ready meals	7.45	3.47	6.25
Snacks	18.77	11.76	17.37
Seasonings	6.96	5.50	7.10
Dairy	9.97	9.11	8.56
Other drinks	4.65	20.10	6.26
Non-fresh fruit and vegetables	3.05	3.80	2.29
Starches	3.47	4.71	2.35
Fresh fruit and vegetables	12.27	24.57	7.12
Soft drinks	3.94	8.95	2.10
All meat	22.75	6.40	36.55
All animal products	39.43	17.14	49.15

Notes: Each column reports the share of the corresponding sample total attributable to each food category. All meat comprises red, white, and other meat. All animal products additionally include fish and dairy. The two aggregate rows are not included in the column totals. Shares may not sum exactly to 100 because of rounding.

3.3 Food Carbon Footprints along the Income Distribution

household food carbon footprints display no economically meaningful income gradient as shown in Figure 3. The correlation between income per consumption unit and annual food emissions per consumption unit is -0.03 (p=0.01). Food carbon footprints are approximately constant across the income distribution, despite substantial differences in food expenditure. This low correlation is robust to conditioning on observable household characteristics. Appendix Table B.3 reports estimates from specifications that include household demographics, and year fixed effects. The estimated income elasticity is 0.01 (p<0.01). This flat relationship also holds within each household size when food carbon footprints and income are measured at the household level rather than per consumption unit (Appendix Figure B.2).

Appendix Table B.4 compares the composition of food baskets across income groups (in the first, fifth, and tenth income deciles). The high-emitting food categories account for similar shares of emissions across the income distribution. Red meat represents 19.4% of emissions in the first income decile, 19.1% in the fifth, and 19.0% in the tenth, and the corresponding shares for all meat are also close (36.4%, 36.5%, and 35.7%). The carbon intensity of these purchases is also very similar across income groups. Within red meat, for example, average emission intensity ranges from 41.30 kg CO_2e/kg in the lowest income group to 42.15 kg CO_2e/kg in the highest. This is despite some differences in the types of meat purchased: higher-income households purchase relatively more whole-muscle cuts and lamb, while lower-income households

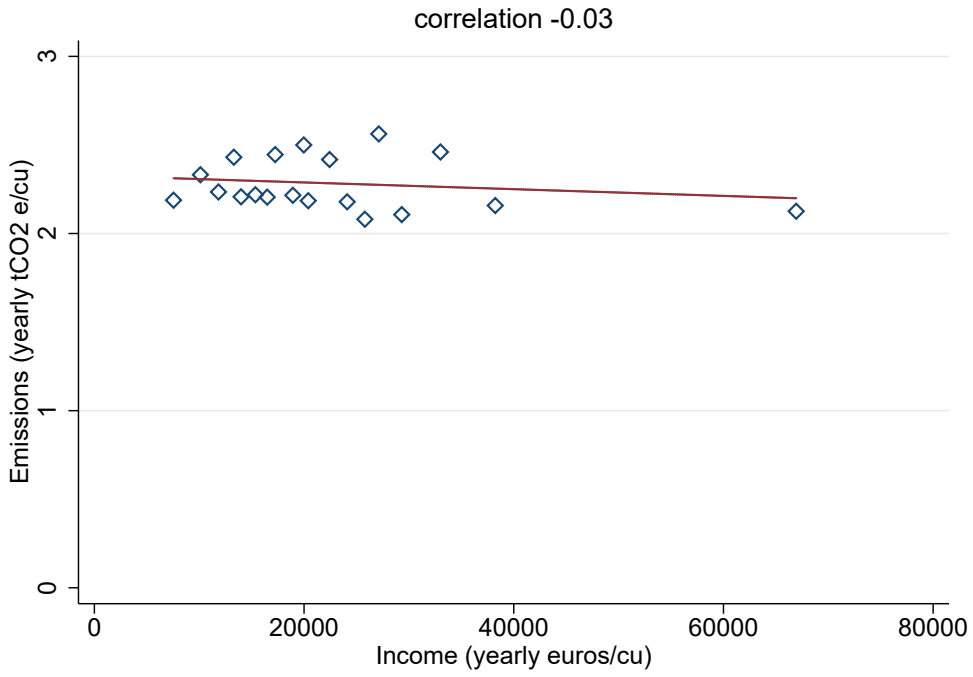


Figure 3: Binscatter Income and Food Carbon Footprints

Notes: Binscatter of annual food carbon footprints against income, both per consumption unit (20 equally sized bins). The pattern holds when excluding the last income bin. Source: Own calculations based on Kantar 2016-2019 and Agribalyse data.

purchase more ground beef. Unit values, by contrast, increase substantially with income. For red meat, the average unit value rises from €9.96/kg in the first income decile to €11.72/kg in the fifth and €15.49/kg in the tenth. Similar gradients are observed for other animal products. These differences suggest substantial within-category differences towards higher-unit-value products as income rises, without a corresponding increase in their average carbon intensity.

Food baskets nevertheless differ across income groups, particularly for categories with much lower emission intensities. The volume share of fresh fruit and vegetables, for example, increases from 16.7% in the first income decile to 30.7% in the tenth, and higher-income households also allocate a larger share of expenditure to fish and smaller shares to snacks, dairy, and soft drinks. Thus, basket composition changes with income, but these changes occur mainly outside the categories that contribute most to food carbon footprints.

Measuring food carbon footprints from expenditure rather than from quantities would lead to a different conclusion. Expenditure-based measures assign a carbon intensity per euro to household spending and may therefore conflate differences in quantities with differences in prices (André et al., 2024). For example, two households may buy the same pack of minced beef, one on promotion and the other at full price. The product, and hence its carbon intensity, is identical, but an expenditure-based measure assigns a larger food carbon footprint to the household that paid more. More generally, prices vary with brand, quality, and store among products with similar carbon intensities. If higher-income households pay more per kilogram for such products, expenditure-based measures overstate their food carbon footprints relative to lower-income households and bias the income gradient upward.

To quantify this measurement difference, I construct an alternative food carbon footprint measure by applying the French carbon intensities per euro from Exiobase3 to household expenditure in the corresponding food categories.[14] Appendix Figure B.4 plots average food carbon footprints by income decile, normalized to one for the lowest decile. Quantity-based food carbon footprints fluctuate between approximately 0.95 and 1.05 and display no systematic relationship with income. In contrast, expenditure-based food carbon footprints rise across the income distribution, reaching 1.35 in the highest decile: the richest households are assigned food carbon footprints approximately 35% larger than the poorest, and the correlation with income is 0.17.

To assess whether the flat gradient reflects the composition of the Kantar sample rather than household food choices, I conduct two additional checks. First, I examine the richest households in the sample. If food carbon footprints increased with income among top earners, footprints should already start rising in the highest income bins of Figure 3. This is not the case: households in the highest bin, with an income of about €67,000 per consumption unit per year, have a food carbon footprint of about 2.1 tCO_2e, close

[14] Exiobase3 classifies products into 200 groups, 25 of which are food or agricultural products, and reports an emission intensity per euro for each group and country. I retain the 14 food groups that households buy directly and that have a counterpart among the food categories in my data, and apply the French intensities to household expenditure in those categories. The note to Figure B.4 lists the retained groups.

to the sample mean of 2.2 tCO_2e. This suggests that the under-representation of top earners is unlikely to hide a strong income gradient. Second, I reproduce the analysis with an individual-level dataset that records all food consumed, including food consumed away from home, for a representative sample of the French adult population.[15] I match the reported quantities to Agribalyse following the procedure described in Berland and Leroutier (2025). INCA3 reports monthly household income using ten ordered brackets, ranging from less than €660 to at least €4,600. Average household income is comparable in the two samples: about €2,780 per month in the Kantar and €2,935 in INCA3.[16] Appendix Figure B.3 plots the weighted mean daily food carbon footprint by income bracket and shows only a modest increase across brackets, from about 8.7 to 9.2 kg CO_2e per day between the lowest and highest brackets. At the individual level, the correlation between income and daily food emissions is 0.06 ($p < 0.01$); conditional on sociodemographic characteristics, the partial correlation falls to 0.03 and is no longer statistically significant ($p = 0.21$).[17]. In INCA3, the share of calories consumed away from home also differs little by income: 30.0% for individuals above the median income and 28.7% for those below (a difference of 1.3 percentage points, $p = 0.05$).[18] The restriction of the Kantar data to food at home is therefore unlikely to mask a large income gradient. Taken together, these results are consistent with the weak income gradient reflecting household food choices rather than the composition of the Kantar sample.

3.4 Other Characteristics of Higher Emitters

Appendix Table B.3 reports the conditional correlations between annual household characteristics and their annual food carbon footprints. Education is negatively correlated with food carbon footprints. Relative to households with primary education or less, households with secondary education have similar food carbon footprints, whereas those with a bachelor's degree have approximately 10% lower footprints and those with a master's degree or above have approximately 15% lower footprints. Food carbon footprints also increase by approximately 0.7% with each additional year in the age of the household head. Conditional on other household characteristics, a household head ten years older is associated with about 7.30% more emissions. Conditional on the other characteristics, rural households have approximately 5.18% higher food carbon footprints, car owners 8.45% higher footprints, and homeowners 2.68% higher footprints.

The age and education gradients are consistent with evidence that younger and more educated individuals consume less meat (Laursen et al., 2019; Koch et al., 2019).

[15] The latest French Individual and National Food Consumption Survey (INCA3), conducted by ANSES in 2014–2015, records the food consumption of 2,121 individuals on a representative day. While it covers food-away-from-home, it is a single cross-section and does not report expenditure. It therefore cannot be used to estimate how households respond to price changes or to simulate the effects of a carbon tax, which require repeated observations of quantities and prices such as those available in the Kantar. I therefore use INCA3 as a complementary check rather than as the main data source.

[16] INCA3 income is imputed using bracket midpoints, half the upper bound for the lowest bracket, and twice the lower bound for the highest bracket. For the Kantar, mean household income is obtained by multiplying mean income per consumption unit (€1,830) by the mean number of consumption units (1.52), which gives about €2,780 per month.

[17] Controls include sex, age group, education, activity status, household size, area of residence, and season fixed effects.

[18] Food-away-from-home includes meals in school and workplace canteens, in restaurants, and at other people's homes.

Using the panel dimension of the data, I examine whether differences in food carbon footprints reflect persistent food-purchasing habits rather than transitory fluctuations. I here restrict the analysis to households observed in consecutive years and classify households above the year-specific 75th percentile as high emitters. Persistence is strong: 74.0% of previous high emitters remain in the top quartile, while only 8.3% of previous non-high emitters enter it. These results indicate that food carbon footprints reflect durable differences in purchased quantities and basket composition, rather than only temporary variation in purchase timing.

Taken together, these results show that income provides little information about a household's position in the food-emissions distribution. Higher emitters are instead distinguished by larger purchased quantities and by demographic characteristics associated with persistent differences in food choices.

3.5 Stated Preferences

To understand how closely environmental concerns align with food choices, and whether stated preferences identify households with lower food carbon footprints, I use a Kantar attitudinal survey matched to households' food purchases and measured food emissions. The survey is available for 2019 only, and the analysis includes 4,744 households.[19] The household head reports agreement with statements about their habits and preferences on a Likert scale from 1 (fully disagree) to 4 (fully agree). I focus on four statements: I try to eat less meat", I try to buy fewer products to avoid waste", I try to preserve the planet's resources", and I try to buy eco-friendly products". Appendix Table B.5 reports summary statistics for responses to each statement. Appendix Figure B.5 shows the associations between these responses and food carbon footprints in 2019. Agreement with "I try to eat less meat" has the strongest negative correlation with measured emissions ($corr = -0.09$, $p < 0.001$), followed by buying eco-friendly products ($corr = -0.06$, $p < 0.001$). The correlations with buying fewer products to avoid waste and preserving the planet's resources are smaller ($corr = -0.05$ and $corr = 0.02$, with $p = 0.042$ and $p = 0.045$, respectively).

Although statistically significant for some questions, these correlations are small. Stated environmental preferences provide limited information about which households have lower food carbon footprints, even when they refer directly to the consumption of the highest emitting food categories. These findings suggest that environmental preferences are only weakly reflected in observed food purchases. They motivate examining how price incentives could encourage less carbon-intensive choices. The next sections estimate food demand and assess how carbon taxation changes the composition of food baskets and the associated emissions.

[19] These households are classified as active and report purchases of random-weight items.

4 Descriptive Evidence on Adjustment Margins

The emissions effects of a food carbon tax depend on whether higher prices lead households to reduce the quantities of high-emission foods they purchase. I document how changes in food prices affect red meat purchases, distinguishing between the decision to purchase and, conditional on purchasing, adjustments in expenditure, quantities, and average unit values.

4.1 Reduced-Form Results

I study how red-meat purchases vary with local prices, distinguishing between the decision to purchase and, conditional on purchasing, adjustments in expenditure, physical quantities, and average unit values. For household i in quarter t, let D_{ijt} indicate whether the household purchases red meat (category j), x_{ijt} denote its total expenditure on that category, and q_{ijt} the quantity purchased. For positive purchases, define the average unit value (i.e., the average price paid per kilo of food in this category) as $v_{ijt} = x_{ijt}/q_{ijt}$. By construction,

$$\ln x_{ijt} = \ln q_{ijt} + \ln v_{ijt}. \tag{1}$$

This identity separates changes in expenditure into changes in kilograms purchased and changes in the average amount paid per kilogram. The unit value is calculated directly from the household's aggregated purchases, without product weights. It can vary because the prices paid vary or because the household changes its mix of cheaper and more expensive products.

I estimate the response of food purchases to price changes along four margins: participation, expenditure, quantity, and unit value. For each outcome, I estimate the following linear regression:

$$Y_{ijt} = \alpha_i + \sum_{j'=1}^{13} \gamma_{jj'} \ln P_{-ij't} + \mathbf{Z}'_{it}\boldsymbol{\delta} + \tau_t + \varepsilon_{ijt}, \qquad Y_{ijt} \in \{D_{ijt}, \ln x_{ijt}, \ln q_{ijt}, \ln v_{ijt}\}. \tag{2}$$

Here, D_{ijt} is an indicator equal to one if household i purchases food category j in quarter t. The participation regression is estimated with a linear probability model (LPM). The coefficient γ_{jj} captures the response to the own-category price index. For participation, a 1% price increase changes the probability of purchasing the category by approximately γ_{jj} percentage points. For the other outcomes, the coefficients measure the percentage change in expenditure, quantity, or unit value associated with a 1% increase in the price index.

The vector $\mathbf{Z}_{it}$ contains household size, age of the household head, rural residence, homeownership, and log income per consumption unit; α_i and τ_t are household and quarter fixed effects. Standard errors are clustered at the household level. The participation regression uses all 94,219 household-quarter observations. The intensive-margin regressions use the 68,973 observations with positive purchases in all

13 categories, matching the demand-system sample.

I use local leave-one-out category price indices as explanatory variables. For each household and quarter, they measure the average price of each food category in the household's *département*, computed from the purchases of all other households in the same département. Départements are the French administrative level between regions and municipalities and correspond to the NUTS 3 level of the European Nomenclature of Territorial Units for Statistics.[20] Unlike the household's own unit values, which are mechanically related to the regression outcomes, these indices exclude the household's own transactions.

Let $\mathcal{K}_j$ denote the standardized products in category j, $\mathcal{H}_{dt}$ the households observed in département d in quarter t, and $d(i)$ the département of household i. The leave-one-out price of product k for household i is the average price paid by the other households i' in its département:

$$p_{-ikt} = \frac{\sum_{i' \in \mathcal{H}_{d(i),t} \setminus \{i\}} x_{i'kt}}{\sum_{i' \in \mathcal{H}_{d(i),t} \setminus \{i\}} q_{i'kt}}, \tag{3}$$

where $x_{i'kt}$ and $q_{i'kt}$ denote the expenditure and quantity of household i' for product k in quarter t. I aggregate these prices into a leave-one-out category price index:

$$\ln P_{-ijt} = \sum_{k \in \mathcal{K}_j} \omega_{kt} \ln p_{-ikt}, \qquad \sum_{k \in \mathcal{K}_j} \omega_{kt} = 1, \tag{4}$$

where ω_{kt} is the national expenditure share of product k within category j in quarter t, common to all households rather than specific to household i's basket.

When no other household in the département purchases product k in quarter t, I omit that product and renormalize the weights over the remaining products. I use this index throughout, in the reduced-form regressions and in the demand system. Since it relies only on other household purchases, it is defined whether or not household i purchases in category j, as the participation regression requires. When the household makes no purchase in the category, the index coincides with the one computed from all households in the département. It is worth noting that these regressions provide descriptive evidence on adjustment margins; they are not interpreted as demand elasticities. I discuss the identifying assumptions for the local price indices in the next section.

Table 2 reports the results. There is no statistically significant association between local red-meat prices and the probability of purchasing red meat (column 1). A 1% increase in the local red-meat price index is associated with a 0.164% decrease in red meat expenditure (column 2). This corresponds to 0.183% lower quantities purchased (column 3) and a small, statistically insignificant 0.019% increase in average unit values (column 4). The reduction in expenditure is therefore primarily associated with

[20] Metropolitan France is divided into 96 départements, with on average about 700,000 inhabitants and 5,700 km^2 each. This dataset covers 95 (Corsica is excluded from the analysis). I use them as local markets because they are small enough to capture geographic variation in prices, yet large enough to contain many purchasing households in each quarter (on average 64 households per département and quarter in the selected sample).

lower quantities purchased. However, the absence of a significant unit-value response does not rule out changes in the composition of red-meat purchases: unit values reflect both the prices paid and the mix of products purchased, so higher product prices may be offset by shifts toward cheaper products. While these regressions provide descriptive evidence on adjustment margins within red meat, they do not establish how households reallocate expenditure across other food categories. I investigate these cross-category adjustments in the next section.

Table 2: Reduced-form responses to red-meat prices

	(1)	(2)	(3)	(4)
Dependent variable	Any purchase	Expenditure	Quantity	Unit value
	D_{ijt}	$\ln x_{ijt}$	$\ln q_{ijt}$	$\ln v_{ijt}$
Red-meat price, $\ln P_{-ijt}$	0.036	-0.164^{**}	-0.183^{***}	0.019
	(0.042)	(0.070)	(0.027)	(0.027)
Prices of other categories	Yes	Yes	Yes	Yes
Household controls	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Household fixed effects	Yes	Yes	Yes	Yes
Observations	94,219	68,973	68,973	68,973

Notes. Each column reports the coefficient on the log price of red meat from a separate regression. The dependent variable in column (1) is an indicator for purchasing red meat. This regression uses the full sample. Columns (2)–(4) are estimated on household-period observations with positive purchases in every food category. Expenditure and quantity are measured per consumption unit; the unit value is expenditure per unit of volume. All regressions include the prices of the other food categories, household and period fixed effects, and controls for rural residence, age of the household head, home ownership, log income per consumption unit, and household size. Standard errors are clustered at the household level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5 Structural Demand Estimation

5.1 Demand Model

The reduced-form results show that higher red-meat prices are associated with lower quantities purchased, with no statistically significant change in purchase participation or average unit values. This raises the question of whether households substitute towards other food categories when relative prices change. This question is particularly relevant for a carbon tax, which would increase the prices of several food categories simultaneously, by different amounts depending on their emission intensity.

I use a continuous demand system to model how households allocate food expenditure across categories. Unlike standard discrete-choice models, which assume that consumers choose one alternative from a set of mutually exclusive options, making all alternatives substitutes by construction, a continuous demand system models the joint allocation of expenditure across several food categories and leaves cross-price

effects unrestricted in sign, so that categories can be either substitutes or complements.[21]

I estimate the Quadratic Almost Ideal Demand System (QUAIDS) of Banks et al. (1997), which extends the AIDS model of Deaton and Muellbauer (1980) by adding a quadratic term in log real food expenditure. This model allows for greater flexibility in the modeling of Engel curves, which describe how budget shares for food categories vary with expenditure.

5.2 Empirical Demand Specification

The model describes how households allocate total food expenditure across $J = 13$ categories as prices and total food expenditure vary. The dependent variable is the budget share allocated by household i to category j in quarter (period) t:

$$s_{ijt} = \frac{x_{ijt}}{X_{it}}, \qquad X_{it} = \sum_{j=1}^{J} x_{ijt}.$$

Let $p_{ijt} = \ln P_{-ijt}$ (the leave-one out price index for category j) and $\mathbf{p}_{it} = (p_{i1t}, \ldots, p_{iJt})$.

The QUAIDS budget-share equation is

$$s_{ijt} = \alpha_j + \sum_{j'=1}^{J} \gamma_{jj'} p_{ij't} + \beta_j \left[\ln X_{it} - a(\mathbf{p}_{it};\theta)\right] + \lambda_j \frac{\left[\ln X_{it} - a(\mathbf{p}_{it};\theta)\right]^2}{b(\mathbf{p}_{it};\theta)} + \varepsilon_{ijt}, \tag{5}$$

where θ denotes the demand parameters. The α_j terms determine the intercepts, and the $\gamma_{jj'}$ terms enter the direct price component of budget shares (own-price for $j = j'$, cross-price otherwise). The β_j terms govern the linear real-expenditure component, while λ_j allows curvature in Engel curves. Note that the $\gamma_{jj'}$ coefficients do not correspond to price elasticities but rather to price effects (as prices also enter real food expenditure).

The two price aggregators are

$$a(\mathbf{p}_{it};\theta) = \alpha_0 + \sum_{j=1}^{J} \alpha_j p_{ijt} + \frac{1}{2} \sum_{j=1}^{J} \sum_{j'=1}^{J} \gamma_{jj'} p_{ijt} p_{ij't}, \tag{6}$$

$$b(\mathbf{p}_{it};\theta) = \exp\left(\sum_{j=1}^{J} \beta_j p_{ijt} \right). \tag{7}$$

The term $\ln X_{it} - a(\mathbf{p}_{it};\theta)$ measures log real food expenditure; $b(\mathbf{p}_{it};\theta)$ scales the quadratic term.

[21] They therefore model the extensive margin (which categories are purchased) rather than the intensive margin (how much is spent on each). The number of alternatives also grows exponentially with the number of goods (2^{13} bundles with 13 categories). In my data, the extensive margin varies little. Most of the variation lies at the intensive margin, which a continuous demand system models directly.

5.3 Identification of Demand Parameters

Food Category Prices. Household transaction prices may be endogenous to purchasing decisions, including product quality, retailer choice, and promotions. As previously discussed, I limit this concern by constructing leave-one-out price indices at the category level using prices paid by other households in the same département. Prices are measured at the detailed product level and aggregated using fixed national expenditure weights, preventing changes in product composition from mechanically affecting the indices.

Identification relies on differences in local price movements across départements over time. Département fixed effects absorb persistent geographical differences in dietary preferences and retail conditions, while quarter fixed effects absorb common national shocks. The variation in prices that is left may reflect differences in distribution costs or in the local pass-through of national cost shocks (such as changes in transportation costs).

The main threat to causal identification is that local price movements may be correlated with unobserved changes in household demand. For example, an increase in local unemployment may reduce red-meat purchases and encourage households to shop at discount retailers. This could lower the average transaction prices observed for the same products, even if retailers do not change their posted prices. Since households within a département are exposed to common economic shocks, the use of leave-one-out price indices does not eliminate this potential source of endogeneity. Controlling for these local shocks is difficult: including département-by-quarter fixed effects absorbs 99% of the variation in the red-meat price index, leaving little variation for identification. An alternative would be to instrument category prices using supply-side cost shocks. However, identifying separate instruments for all 13 food categories is challenging because many categories share production inputs and are exposed to similar cost shocks. I therefore rely on local price variation, conditional on the observed controls and fixed effects, and assess the sensitivity of the estimates to alternative specifications in Section 5.5.

Food Expenditure In equation (5), total food expenditure enters both the denominator of the dependent variable, $s_{ijt} = x_{ijt}/X_{it}$, and the real-expenditure terms on the right-hand side. More generally, unobserved demand shocks may affect both the allocation of expenditure across categories and total food expenditure. I therefore treat $\ln X_{it}$ as endogenous and instrument it with annual household income, following Banks et al. (1997). To better account for household living standards, I divide income by the number of consumption unit in the household. The identifying assumption is that, conditional on total food expenditure, prices, and household characteristics such as age and education, income per consumption unit does not independently affect the allocation of the food budget.

This assumption may be demanding for categories with income-related preferences such as fresh fruits and vegetables, snacks or fish. For meat, however, I find no clear income gradient in meat budget shares. Appendix Table B.4 presents these descriptive results. Red meat accounts for 5.51%, 5.62%, and 5.65%

of food expenditure in the first, fifth, and tenth income deciles, respectively. The corresponding shares for white meat and other meat are also relatively similar across income groups. While these patterns do not establish the exclusion restriction, they support its plausibility for the meat categories, which are particularly important for the analysis as they emit the most. The first-stage F-statistic for the excluded instrument is 154.

5.4 Estimation

The demand system is estimated using the iterated linear least-squares (ILLS) estimator proposed by Lecocq and Robin (2015). When total food expenditure is treated as endogenous, the first-stage residual is included in each budget-share equation. The system is then estimated by iterating a series of seemingly unrelated regressions and updating the nonlinear price aggregators until convergence.

The budget-share equations are estimated jointly, imposing the adding-up, homogeneity, and symmetry restrictions implied by consumer theory. I set $\alpha_0 = 0$ prior to estimation. The Stone index, calculated using the sample average budget shares, is used only to initialize this iterative procedure. The estimation is restricted to household-period observations with positive budget shares in every food category.[22]

I use the estimated parameters to compute own-price, cross-price, and expenditure elasticities. Appendix C.1 presents the corresponding formulas. Price elasticities account for both the direct price effects and the effects operating through the price aggregators and real food expenditure. Unlike the $\gamma_{jj'}$ parameters, they measure the full model-implied quantity response to price changes. Marshallian price elasticities are calculated holding nominal total food expenditure constant.

5.5 Demand estimates and elasticities

Table 3 reports the predicted budget shares and the food-expenditure and own-price elasticities evaluated at the full-sample average. Column (1) presents the predicted budget shares, column (2) the food-expenditure elasticities, and columns (3) and (4) the uncompensated and compensated own-price elasticities. The predicted budget shares are close to the observed average shares across categories, indicating that the model reproduces the average composition of household food expenditure well. Appendix Table C.2 reports the full matrix of compensated price elasticities.

Fresh fruit and vegetables and fish are the most elastic to additional food expenditure, with elasticities of 2.239 and 1.956, respectively. Red meat is the most price-elastic category, with an uncompensated own-price elasticity of -1.465 and a compensated elasticity of -1.410. This suggests that a price policy targeting red meat, the most emission-intensive category, could lead to a substantial reduction in its

[22] To the best of my knowledge, there is not yet a consensus on how to account for censoring while maintaining consistency with utility theory; see Amano-Patiño (2019).

demand. The magnitudes of the price and expenditure elasticities are close to those reported by Amano-Patiño (2019) for the United States and by Caillavet et al. (2016) for France.[23]

These price elasticities are derived from the estimated budget-share equations rather than directly estimated from quantities. They account for both the change in expenditure allocated to a category and the direct effect of its price on the quantity that can be purchased. For red meat, which has a predicted budget share of 6.8%, a 1% price increase reduces its budget share by approximately 0.032 percentage points. Combined with the direct price effect, this corresponds to an uncompensated own-price elasticity of -1.465. Appendix C.1 provides the formulas used to recover the elasticities from the estimated demand parameters (see equation C.4). The own price elasticity is larger in magnitude than the reduced-form coefficient for red-meat quantities (-0.183) reported in Section 4. The two estimates are not directly comparable: the reduced-form regression directly estimates changes in kilograms of red meat purchased, whereas the QUAIDS elasticity is derived from estimated changes in budget shares, holding total food expenditure constant. Moreover, the demand system jointly models expenditure allocation across all 13 food categories, while the reduced-form regression focuses on within red meat adjustments.

To examine how households reallocate their food expenditure when the price of red meat changes, I use the compensated cross-price elasticities reported in Appendix Table C.2. A 1% increase in the price of red meat increases demand for starches by 0.298 percent, dairy by 0.188%, snacks by 0.210%, ready meals by 0.143%, soft drinks by 0.124%, seasonings by 0.186%, and non-fresh fruit and vegetables by 0.153%. Demand for other meat also increases by 0.093%. These statistically significant positive elasticities indicate substitution away from red meat toward these categories. By contrast, fresh fruit and vegetables and fish have negative cross-price elasticities of -0.261 and -0.205, respectively, suggesting that they are complements to red meat. The elasticity for white meat is positive but statistically insignificant, providing no evidence that households substitute from red meat toward white meat when the price of red meat increases.

As a specification check, I estimate separate 2SLS budget-share regressions using the same 13 category-level leave-one-out price indices and household controls as in the linear version of the AIDS model, with département and quarter fixed effects. Log food expenditure is instrumented using household income. Unlike the AIDS model, these regressions do not impose symmetry, homogeneity, or adding-up restrictions. Appendix Section C.4 presents the estimating equation, and Appendix Table C.3 compares the estimated own-price coefficients. For red meat, the estimated coefficient on the red-meat price is -0.034, close to the corresponding AIDS parameter $\gamma_{33} = -0.029$. Estimates are also broadly similar for several other categories, suggesting that the estimated price coefficients are not very sensitive to the restrictions imposed by the demand system.

I then use this simpler specification to investigate the sensitivity of the price coefficient to alternative fixed effects that are computationally difficult to accommodate in the full demand system. First,

[23]For a detailed comparison of the estimates, particularly for meat and dairy products, see Appendix Table C.1.

I add département-specific linear trends to the baseline département and quarter fixed effects. While département fixed effects absorb persistent local differences and quarter fixed effects absorb common national shocks, these trends additionally account for gradual changes specific to each département, such as a progressive shift away from red-meat consumption. The coefficient changes from -0.030 to -0.039. However, these trends do not capture temporary local shocks. I next examine the role of persistent unobserved household heterogeneity. Household fixed effects cannot be incorporated into the full demand system because of computational constraints associated with its dimensionality. I therefore replace département fixed effects with household fixed effects in the simpler budget-share regression. These absorb time-invariant differences in household preferences and purchasing behaviour. The coefficient decreases to -0.007, or -0.013 when region-by-quarter fixed effects are also included. Overall, the estimated red-meat price coefficient remains negative and statistically significant across specifications. Its magnitude is relatively stable when accounting for département-specific trends and regional shocks, but decreases substantially with household fixed effects.[24] While these checks address several potential sources of confounding, they cannot rule out time-varying local demand shocks correlated with prices. The causal interpretation of the demand estimates therefore remains conditional on the exogeneity of the remaining local price variation.

5.5.1 Heterogeneity in Reactions Across Income Groups

I study heterogeneity in demand responses by estimating the demand system separately by income quartile. Overall, own-price elasticities do not exhibit a systematic income gradient. For many categories, including white meat, fish, and non-fresh fruit and vegetables, estimated elasticities are broadly similar across income groups. A clearer income gradient emerges for soft drinks, for which demand becomes more price-responsive as income rises. The own-price elasticity increases in magnitude from -0.265 in the bottom quartile (not statistically significant) to -1.341 in the top quartile. For red meat, differences are more modest: the estimated elasticity is -1.244 in the bottom quartile, compared with -1.508, -1.659, and -1.543 in the subsequent quartiles. This suggests that lower-income households may be somewhat less responsive to changes in red-meat prices, although the relationship is not monotonic.

Expenditure elasticities also vary across income groups, but do not exhibit a clear income gradient. The expenditure elasticity of fresh fruit and vegetables is 1.75 in the bottom quartile, falls to 1.31 and 0.99 in the middle quartiles, and rises again to 1.64 in the top quartile. Some categories exhibit more systematic differences. For non-fresh fruit and vegetables, the expenditure elasticity is highest in the bottom quartile (1.204), compared with 0.711–0.829 in the other quartiles. Overall, these results point to some category-specific heterogeneity in demand responses, but no systematic differences in price or expenditure elasticities across income groups.

[24] The lower estimated price coefficient decreases does not imply a correspondingly small price elasticity. At a red-meat budget share of approximately 6–7%, the coefficient translates into an approximate own-price elasticity of -1.1, holding nominal food expenditure constant.

Table 3: Predicted shares, food-expenditure elasticities, and uncompensated and compensated own-price elasticities

	Predicted share	Expenditure elasticity	Uncompensated own-price elasticity	Compensated own-price elasticity
Starches	0.038***	0.967***	-1.252***	-1.216***
	(0.000)	(0.030)	(0.053)	(0.053)
Fresh fruit/veg.	0.081***	2.239***	-0.460***	-0.279***
	(0.004)	(0.107)	(0.083)	(0.084)
Red meat	0.068***	0.808***	-1.465***	-1.410***
	(0.003)	(0.035)	(0.058)	(0.058)
White meat	0.087***	0.784***	-1.141***	-1.073***
	(0.003)	(0.030)	(0.056)	(0.056)
Other meat	0.107***	0.782***	-0.921***	-0.837***
	(0.002)	(0.023)	(0.047)	(0.047)
Fish	0.049***	1.956***	-1.018***	-0.923***
	(0.002)	(0.076)	(0.083)	(0.083)
Dairy	0.098***	0.909***	-1.196***	-1.107***
	(0.002)	(0.024)	(0.068)	(0.068)
Snacks	0.197***	0.824***	-0.796***	-0.634***
	(0.002)	(0.019)	(0.053)	(0.053)
Ready meals	0.075***	0.859***	-1.151***	-1.087***
	(0.001)	(0.041)	(0.077)	(0.077)
Soft drinks	0.048***	0.762***	-0.859***	-0.823***
	(0.001)	(0.036)	(0.063)	(0.063)
Seasonings	0.077***	0.649***	-0.923***	-0.874***
	(0.001)	(0.021)	(0.024)	(0.024)
Other drinks	0.042***	1.189***	-1.233***	-1.183***
	(0.001)	(0.044)	(0.044)	(0.044)
Non-fresh fruit/veg.	0.033***	0.832***	-0.953***	-0.925***
	(0.001)	(0.034)	(0.019)	(0.019)

Notes: Predicted budget shares and elasticities are evaluated at the full-sample average. The predicted share is the proportion of total food expenditure allocated to each category; for example, 0.068 corresponds to 6.8 percent. The expenditure elasticity gives the percentage change in demand for a category associated with a 1%increase in total food expenditure. The own-price elasticity gives the percentage change in demand associated with a 1%increase in the price of the same category. Uncompensated elasticities include both substitution and income effects, whereas compensated elasticities hold utility constant and isolate the substitution effect. Standard errors calculated with the delta method are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

6 Policy Simulations

I use the estimated demand system parameters to simulate the effects of extending the current French carbon price to food products. I first consider a tax applied to all food products and then compare it with narrower tax bases covering all meat products or red meat only.

6.1 Policy Design and Simulation

Carbon price and taxation scenarios. I consider a consumer-level carbon tax proportional to the life-cycle carbon intensity of food products.[25] The carbon price is set at €44.60 per tCO_2e, which is the rate of the carbon component of French energy taxes.[26]

I compare three taxation scenarios. The benchmark scenario applies the carbon tax to all food products. The second design restricts the tax to meat products, while the third applies it only to red meat. The carbon price per ton of CO_2e is the same under all three designs. The policies therefore differ only in the products included in the tax base.

For household i and category j, the tax-induced price increase, expressed in euros per kilogram of food, is

$$\Delta p_{ij} = \tau_{\text{CO2}} \times eij, \tag{8}$$

where $\tau_{CO_2} = 0.0446$ €/kgCO2e and eij is the average carbon intensity of household i's purchases in category j, measured in $kgCO_2e$ per kilogram of food. This intensity is calculated by dividing the total emissions associated with the household's purchases in a category by the total quantity purchased. Thus, even within the same food category, households face different tax-induced price increases depending on the composition of their initial purchases.

I hold e_{ij} fixed in the simulations. Behavioral responses therefore operate through changes in quantities and expenditure across food categories, without capturing substitution toward less carbon-intensive products within a category.

Figure 4 presents the simulated percentage price increases for the main food categories under the benchmark scenario. These increases depend on both carbon intensity and initial prices. Red meat experiences the largest average increase, approximately 17%, compared with approximately 5% for white

[25] A consumer-level tax avoids carbon leakage, i.e. the risk that foreign-produced goods become relatively cheaper than domestically produced ones, inducing a shift in demand towards imports.

[26] The French *contribution climat-énergie* was introduced in 2014 at €7 per tCO_2 and rose to €44.60 per tCO_2 in 2018, a level at which it has since remained. It is a carbon component incorporated into taxes on fossil fuels, including petrol, diesel, natural gas, and heating oil. Food products are not, however, taxed according to their carbon footprint.

meat. The figure also reports the 10th and 90th percentiles, reflecting differences across households in the carbon intensity and initial prices of their category-level purchases.

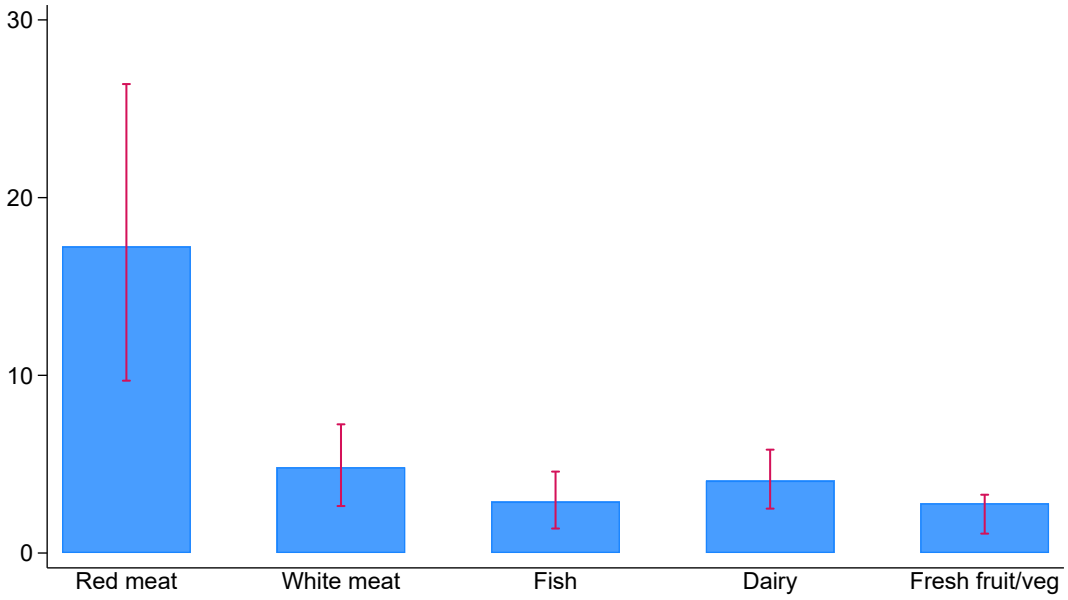


Figure 4: Simulated price variation following a carbon tax.

Notes: Bars report the average percentage increase in household-specific category prices. The capped lines indicate the 10th and 90th percentiles. Dispersion within categories reflects differences in the composition of products initially purchased across households. For readability, only five food categories are shown.

Interpretation and scope of the counterfactual. The demand system characterizes how households adjust their purchases to the retail prices they face, but it does not model how a tax is transmitted through the food supply chain. In the simulations, I assume that retailers leave their existing prices unchanged and that the carbon tax is added on top. Consumers therefore face the full tax as a price increase. This full-pass-through assumption provides a partial-equilibrium benchmark; it is not a claim that competition would necessarily generate full pass-through. In other words, the counterfactual scenario studies how households would change the composition of their food baskets if the same carbon price per ton of CO_2e were fully reflected in the food prices they face when shopping.

If pass-through was incomplete but uniform across food categories, it would be equivalent to applying a lower carbon tax on food. It would scale down all simulated price increases proportionally while preserving their ranking across categories. Different pass-through rates across categories could instead alter relative price changes and, consequently, household substitution patterns.[27] Modeling retailers' and producers' pricing decisions is outside the scope of this paper. Evidence from soda taxes nevertheless shows that full

[27] Under full pass-through, average prices increase by approximately 17% for red meat and 4% for white meat. Let ρ_R and ρ_W denote their respective pass-through rates. Equalizing the two price increases would require $0.17\rho_R = 0.04\rho_W$, or $\rho_R/\rho_W \simeq 0.24$. Red-meat pass-through would therefore need to be approximately one quarter of white-meat pass-through.

pass-through can provide a plausible benchmark in some food markets.[28]

Over time producers and retailers may respond to changes in demand by adjusting production technologies, prices and product portfolio. The counterfactual holds these supply-side margins fixed, the estimates thus capture the short-run household response rather than the full long-run equilibrium effect of the tax.

The simulations hold nominal food expenditure constant and therefore capture demand responses conditional on the total food budget. Revenue redistribution is incorporated into the incidence analysis but not into households' subsequent consumption decisions. Nevertheless, the estimated expenditure elasticities for meat products are relatively low (around 0.8), suggesting that additional food expenditure from rebates would lead to less-than-proportional increases in meat consumption.

6.2 Effects on Food Carbon Footprints

The annual food carbon footprint in the estimation sample averages approximately 2.2 tCO_2e per consumption unit. Under the assumption of constant nominal food expenditure, applying the carbon tax to all food products reduces aggregate food carbon footprints by 7.90% and raises the food price index by 4.61%.

This magnitude is close to the 7.5% reduction reported by Caillavet et al. (2016), but is obtained at a lower cost to households: their scenario requires a flat 20% ad valorem increase on eight animal-based food groups, against a 4.61% increase in the price index here. The difference lies in the design of the tax instrument. The uniform ad valorem rate they use raises the price of every taxed good in the same proportion, so that relative prices within the taxed set are left unchanged: a household switching from beef to pork, or from beef to poultry, faces the same trade-off as before the tax. An emissions-based tax instead accounts for differences in emission intensities: the tax burden is aligned with the environmental damage.

The measured reduction in emissions is higher than in Bonnet et al. (2018) where they obtain only a 1.90% decrease in food carbon footprints after taxing all animal products at €56/tCO_2e, a tax level close to mine. Two features of their specification account for this. First, the use of a different structural demand model (random coefficient logit) restricts cross-price effects to be non-negative, so complementarities between taxed animal products and untaxed but emission-intensive goods cannot be represented. In addition, the simulations hold the total quantity of food purchases fixed which does not allow total food quantities to decrease in response to higher prices. In this paper, I instead hold nominal food expenditure constant, so that the rise in the food price index reduces real purchasing power. Part of the reduction

[28] Dubois et al. (2020) find pass-through close to 100% for two major soft-drink brands following the introduction of the UK soda tax. However, this evidence concerns the one market and does not establish uniform pass-through across the broader food sector.

obtained here therefore comes from a contraction in total food quantities rather than from recomposition of the food basket alone.

Narrowing the tax base reduces both the change in food carbon footprints and the increase in the food price index. Taxing all meat products reduces food carbon footprints by 5.04% and raises the food price index by 1.70%. Restricting the tax to red meat reduces food carbon footprints by 4.25%, while increasing the food price index by 0.92%.

The meat tax therefore retains approximately 64% of the reduction obtained under the benchmark scenario while generating 37% of its food-price increase. The red-meat tax retains approximately 54% of the benchmark reduction while generating only 20% of its food-price increase. These results highlight the central contribution of red meat to the response of food carbon footprints.

6.3 Distributional Effects

Incidence measures and revenue recycling. I distinguish between tax payments and the welfare cost of the tax-induced price change. Let q_{ij}^0 and q_{ij}^1 denote household i's baseline and post-tax quarterly quantities per consumption unit in category j. Annual tax payments per consumption unit are calculated using post-tax quantities:

$$T_i = 4 \sum_j \Delta p_{ij} q_{ij}^1. \tag{9}$$

I approximate the welfare cost of the price change by evaluating it at baseline quantities:

$$\widehat{CV}_i = 4 \sum_j \Delta p_{ij} q_{ij}^0. \tag{10}$$

Because this measure holds quantities at their baseline levels, it is a first-order approximation to compensating variation rather than an exact compensating-variation measure.

I consider a revenue-neutral scheme that returns aggregate tax revenue uniformly across consumption units. Let s_i denote the number of consumption units in household i and w_i its survey weight. The rebate per consumption unit is

$$r = \frac{\sum_i w_i s_i T_i}{\sum_i w_i s_i}. \tag{11}$$

Let y_i denote annual income per consumption unit. The gross tax payment is T_i/y_i, while the net tax

and welfare burdens are $(T_i - r)/y_i$ and $(\widehat{CV}_i - r)/y_i$, respectively.

Incidence before revenue recycling. Annual tax payments per consumption unit vary comparatively little across the income distribution, consistent with the weak relationship between income and food carbon footprints. They amount to €92.44 in the bottom income decile and €88.30 in the top decile. Payments are not monotonic in income and remain within a relatively narrow range across all ten deciles.

The same payments represent very different shares of household income. The gross tax payment declines from 1.13% of income in the bottom decile to 0.20% in the top decile. The proportional burden is therefore about 5.7 times larger at the bottom than at the top of the income distribution.

The Kakwani index confirms this result. It compares the concentration coefficient of tax payments, when households are ranked by pre-tax income, with the Gini coefficient of income.[29] The concentration coefficient of tax payments is 0.031, compared with an income Gini coefficient of 0.276, producing a Kakwani index of −0.245. The tax is regressive because payments vary much less across households than income: relatively similar payments in euros represent a considerably larger share of income for poorer households.

Incidence after revenue recycling. The benchmark tax generates a uniform rebate of approximately €90 per consumption unit.[30] Since mean annual payments range from €88.30 to €96.05 per consumption unit across income deciles, average net tax payments remain close to zero in nominal terms throughout the income distribution. The rebate does not, however, fully offset the welfare cost of the price increase. Mean net first-order compensating variation is positive in every income decile and decreases along the income distribution. It represents 0.49% of income in the bottom decile and 0.07% in the top decile. Equal recycling therefore largely offsets differences in tax payments but does not fully compensate households for the welfare cost of the price change.

Incidence within income deciles. Households with similar incomes can face different net welfare burdens because the carbon content and composition of their food baskets differ. This is the horizontal dimension of tax incidence: income identifies the household's position in the income distribution but does not fully predict its exposure to the tax. Dispersion in net welfare burdens is greatest at the bottom of the income distribution. The within-decile interquartile range declines from 0.515 percentage points in the bottom decile to 0.102 percentage points in the top decile. Poorer households therefore experience both higher average welfare losses and greater dispersion in those losses relative to income. Part of this pattern is mechanical, since a given difference in euro losses represents a larger share of income for poorer households. It nevertheless shows that income alone is insufficient to characterize the incidence of the

[29] A negative Kakwani index indicates that the tax is regressive relative to income.

[30] Since rebates are proportional to household consumption units, larger households receive larger transfers, while the rebate per consumption unit is identical across households.

tax: households with similar incomes can face different burdens because their food baskets expose them differently to carbon pricing.

6.4 Distributional Effects of Alternative Tax Designs

Figure 5 compares gross tax payments across income deciles under the three tax designs. Panel (a) reports annual payments per consumption unit, while Panel (b) expresses these payments as a share of annual income. In all three scenarios, payments in euros vary comparatively little across the income distribution, whereas the proportional burden declines sharply with income. In the bottom income decile, gross tax payments represent 1.13% of income under the all-food tax, 0.44% under the all-meat tax, and 0.24% under the red-meat-only tax. The corresponding burdens in the top decile are 0.20%, 0.08%, and 0.04%. Despite these differences in magnitude, the three tax bases display very similar distributional patterns. For each design, the proportional burden in the bottom decile is approximately six times that in the top decile. Narrowing the tax base therefore reduces the tax payment but does not eliminate its regressivity, because tax exposure is weakly related to income.

Net welfare burdens follow the same ordering: the all-food tax generates the largest welfare effects, followed by the all-meat and red-meat-only taxes. However, narrower tax bases also deliver smaller emissions reductions. The all-food tax increases the food-price index by 4.61% and reduces emissions by 7.90%. The all-meat tax achieves a 5.04% reduction—64% of the reduction under the all-food tax—with a 1.70% food-price increase, or 37% of the increase under the all-food tax. The red-meat-only tax reduces emissions by 4.25%—54% of the benchmark reduction—while increasing the food-price index by only 0.92%, or 20% of the benchmark increase. Targeting meat, and red meat in particular, retains a substantial share of the emissions reduction while considerably limiting food-price increases and household payments.

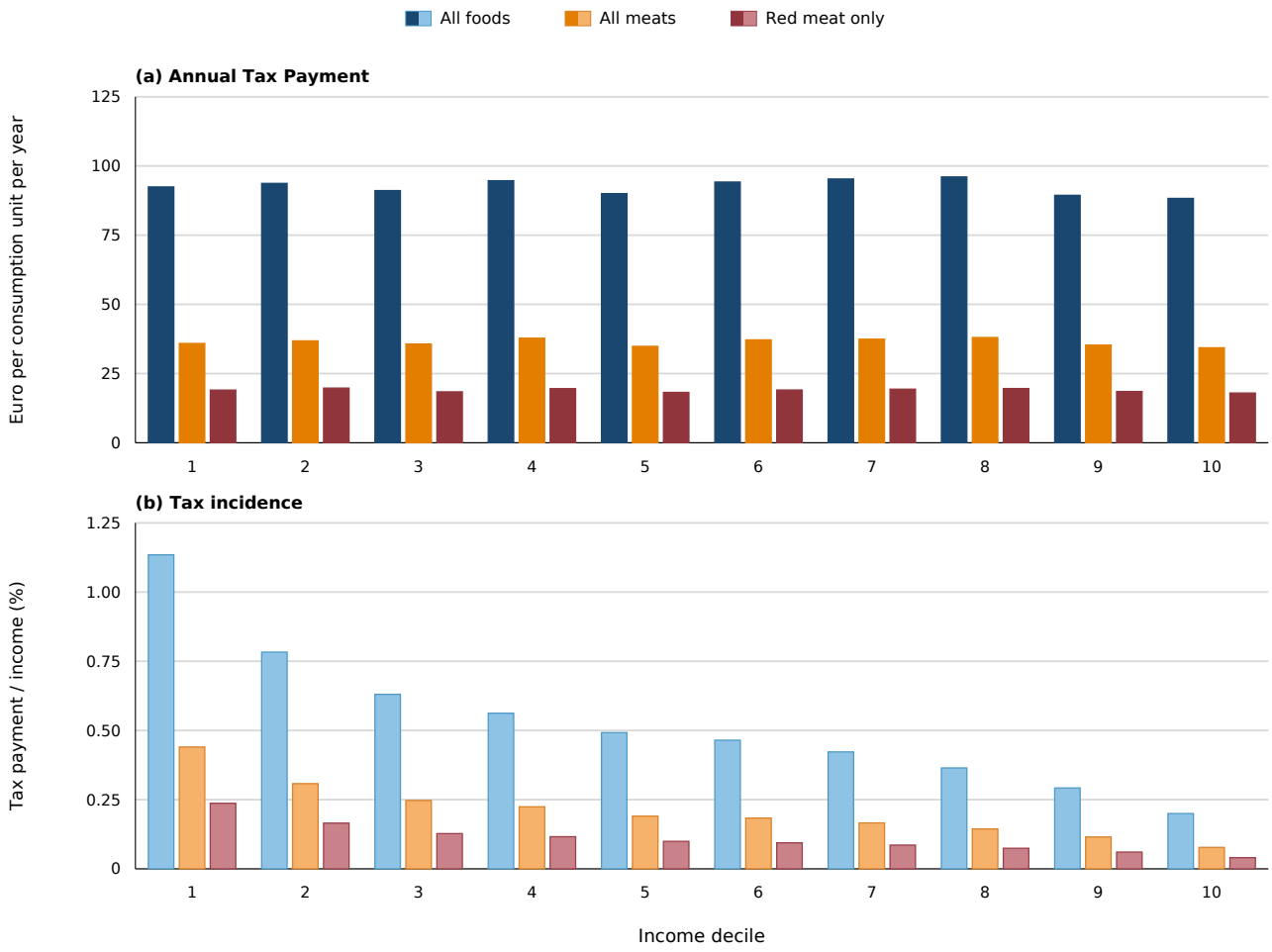


Figure 5: Tax Payments and Tax Incidence across Income Deciles under Alternative Tax Designs

Notes: Panel (a) reports mean annual tax payments in euros per consumption unit. Panel (b) reports annual tax payments as a percentage of annual income. Bars show averages within income deciles, numbered from the lowest to the highest income decile. Tax payments are calculated using simulated post-tax quantities and annualized by multiplying quarterly payments by four. The three scenarios tax all food products (the benchmark), all meat products, or red meat only. Both panels report gross tax payments before revenue recycling.

7 Conclusion

This paper studies the potential of food carbon taxation to reduce emissions and its distributional consequences in France. To do so, I construct a fine-grained dataset by matching detailed household food purchases to product-level emission intensities. I find that food carbon footprints are weakly correlated with income, with high-emitting households found throughout the income distribution. Other household characteristics do not clearly identify distinct groups of high emitters either. Red meat consumption plays an important role in food carbon footprints: while accounting for only 1.3% of food purchases in volume, it represents 19.1% of food emissions and contributes substantially to differences across households. These stylized facts provide a rationale for studying a food carbon tax, which targets high-emission consumption directly by increasing the relative price of carbon-intensive products.

The effectiveness of a food carbon tax depends on how households respond to price changes. Reduced-form estimates show that households respond to higher red-meat prices primarily by reducing purchased quantities, rather than withdrawing from the category. A structural demand model further shows that red meat is the most price-elastic food category. Based on these estimates, I simulate a carbon tax of €44.6 per tCO_2e. Taxing all food products could reduce food carbon footprints by 7.9%, while increasing the food price index by 4.6%. I also examine alternative tax designs: restricting the tax to meat reduces emissions by 5.0%, while taxing red meat alone delivers a 4.3% reduction, with smaller increases in food prices. Average tax payments are similar in absolute terms across income deciles, but represent a substantially larger share of income for poorer households, making the tax regressive. Redistributing tax revenues uniformly per consumption unit brings average net tax payments close to zero in each income decile, although it does not fully compensate households for the welfare cost of higher food prices.

These results highlight an important trade-off in the design of food carbon taxes. Narrowing the tax base to red meat alone retains approximately 54% of the emissions reduction achieved by taxing all foods, while generating only 20% of the increase in the food-price index. The choice of tax base affects the magnitude of emissions reductions and household payments, while revenue recycling plays a central role in determining the distributional consequences of the policy. This paper's results inform the potential effectiveness and distributional consequences of food carbon taxes. The political acceptability of this policy also depends on factors such as cultural preferences, and the broader political context that are beyond the scope of this analysis.

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Appendix

A Data Appendix

A.1 Kantar WorldPanel Representativeness

Table A.1: Representativeness of the Kantar WorldPanel sample

	Kantar sample		
Variable	Full sample	Selected sample	National benchmark
Income per CU (€/month)	1,785	1,830	1,977
Age of household head (years)	47.82	52.27	53.0
Household size (persons)	2.72	2.19	2.20
Share college educated (%)	45.6	41.0	40.0
Observations	59,296	24,152	–

Notes: The table compares moments from the Kantar household panel with national benchmarks from INSEE and related official statistical sources. The Kantar WorldPanel full-sample moments are unweighted. The selected-sample moments are weighted using the annual weights provided by Kantar WorldPanel. The selected sample corresponds to active households reporting random-weight items. National benchmarks refer to 2016–2019 averages when available; age, household size, and education are benchmarked using 2018 statistics. Income is measured in constant 2015 euros per consumption unit. The Kantar income variable is self-reported in brackets, whereas the INSEE benchmark uses disposable income from the Tax and Social Income Survey after taxes and transfers. The college-educated share is defined as the share with Bachelor or Master-level education.

Table A.2: Food at Home: Total Spending and Category Budget Shares, Kantar WorldPanel vs. Consumer Expenditure Survey.

	Kantar WorldPanel 2016–2019	Household Budget Survey 2017	Difference
Total food at home, excl. alcohol (constant 2015 € / CU / month)			
Food at home	199.59	219.10	+9.8%
Budget shares (% of food at home, pp difference)			
Meat	20.6	20.5	−0.1
Fish	6.3	5.9	−0.4
Dairy	12.8	12.3	−0.5
Fruits & vegetables	15.4	14.5	−0.9
Soft/sugary drinks	3.8	3.8	+0.0

Notes: Comparison between Kantar WorldPanel and Consumer Expenditure Survey (*Budget de Famille*). Total food at home is mean spending per consumption unit, in euros per month, household-weighted, and *excludes alcoholic beverages*. All monetary amounts are in *constant 2015 euros*: Kantar WorldPanel is deflated by the CPI for consumer goods (base 2015) and Budget de Famille 2017 by the INSEE all-household CPI (factor 100/101.2). Its difference is the percentage gap (Budget de Famille−Kantar WorldPanel)/Kantar WorldPanel. Budget shares are the household-weighted mean of household-level expenditure ratios within the alcohol-excluded food-at-home basket; Kantar WorldPanel shares are renormalised to that basket. Shares are scale-invariant, so the deflation does not affect them, and their Difference is the percentage-point gap (Budget de Famille−Kantar WorldPanel). Kantar WorldPanel is a homescan panel and does not observe food away from home or total food expenditure.

A.2 Linking Food Transactions with Environmental Indicators

A.2.1 Matching Procedure

The unit of the matching procedure is a Kantar product identifier. The raw data contain between 191,476 and 204,891 distinct product identifiers annually over 2016–2019, covering more than 18 million transactions per year. The product identifiers cannot be linked directly across years because products enter and exit the market and Kantar assigns new identifiers when their format or composition changes. Product attributes, such as the type of meat, fat content, flavor, package size, organic status, and private-label status, are more stable and can therefore be used to identify corresponding products.

To calculate food carbon footprints, I match each Kantar WorldPanel product to a standardized product in Agribalyse and assign the corresponding emission factor. Kantar WorldPanel records approximately 200,000 distinct product identifiers each year and provides detailed information on their brands and attributes. I first use a matching table developed by NutriXConso, a previous effort that linked food purchases to nutritional outcomes. This table links earlier Kantar WorldPanel product identifiers to the same standardized food products used in Agribalyse (De Mouzon and Orozco, 2011). I can therefore use these existing matches for the carbon-footprint analysis, and I supplement them with manual matches.

Because Kantar WorldPanel identifiers change over time and new products appear, many products observed in 2016–2019 remain unmatched. I therefore train a random-forest classifier on 105,831 products with existing or manual matches in 2011. Using Kantar product attributes as predictors, the classifier distinguishes among the 939 standardized products represented in the training data and assigns matches to 569,295 previously unmatched product-year records.

The classifier assigns matches to 129,016 products in 2016, 139,306 in 2017, 147,747 in 2018, and 153,226 in 2019, for a total of 569,295 product-year observations. Once the pre-existing and manual matches are added, the final data contain between 1,047 and 1,062 distinct Agribalyse standardized products annually. This number exceeds the 939 outcomes available to the classifier because manual matching introduces additional standardized products. Together, the three matching stages assign emission factors to Kantar products accounting for 98% of transactions over the study period.

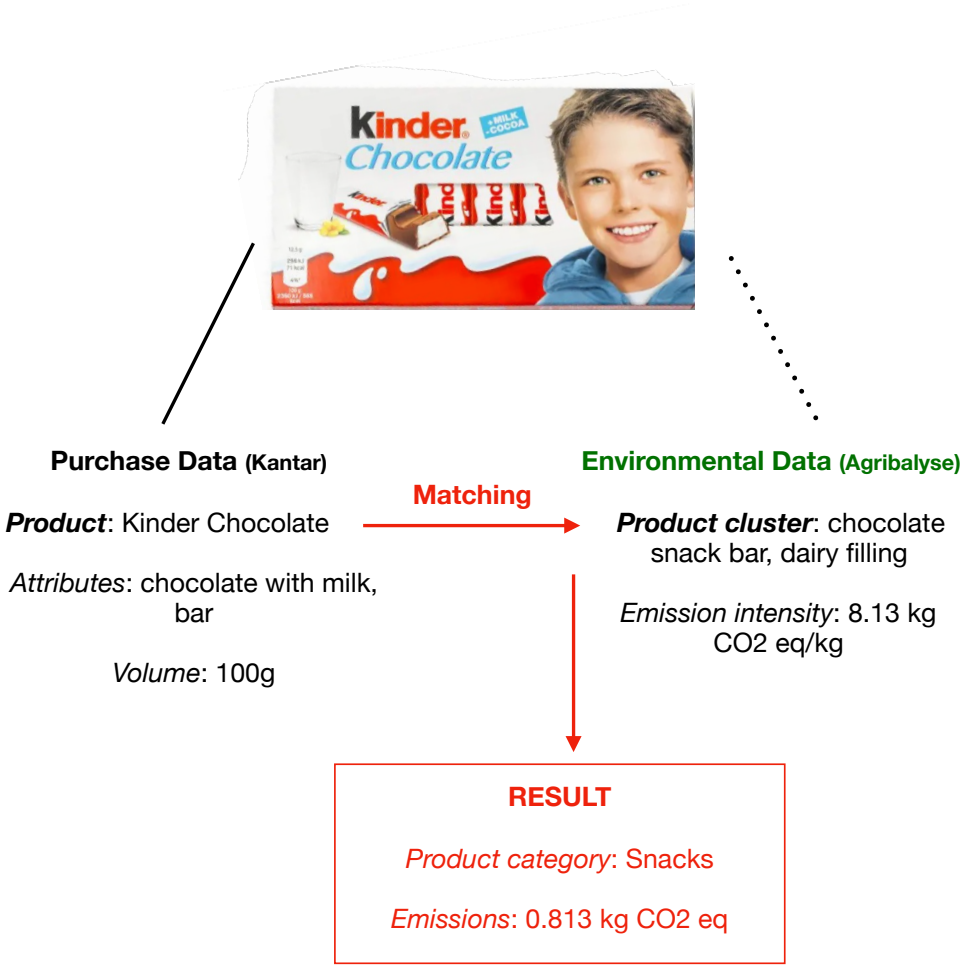


Figure A.1: Matching Example in the Snacks Category.

Notes: Simple example taken from the real-life setting.
The variable and product names are not extracts from the Kantar WorldPanel dataset.

A.2.2 Validation

I evaluate the classifier using products with existing or manual matches in 2016–2019 as ground truth. For these products, I conceal the known match, predict it using the classifier trained on the 2011 sample, and compare the prediction with the existing assignment. For validation, I retain only products whose known Agribalyse match is among the outcomes learned by the classifier from the 2011 data.

Table A.3 reports the matching performance by year. Accuracy is the proportion of products assigned to exactly the correct Agribalyse standardized product. Macro-F1 combines precision and recall for each standardized product and then gives equal weight to all standardized products, irrespective of how frequently they occur. Accuracy ranges from 92.3% to 93.5%, while macro-F1 ranges from 0.820 to 0.839, indicating that performance is less uniform across standardized products and is weaker for some less frequently represented outcomes.

Table A.3: Random-Forest Validation by Year

Year	Products evaluated	Accuracy	Macro-F1
2016	55,211	0.935	0.839
2017	50,809	0.934	0.838
2018	47,088	0.927	0.826
2019	43,500	0.923	0.820

Notes: The validation sample contains products with existing or manual matches whose Agribalyse standardized product is represented in the 2011 training data. Accuracy is the exact-match rate. Macro-F1 gives equal weight to performance for each standardized product.

To identify where exact-product errors are concentrated, I pool the annual validation samples and calculate exact-match accuracy within detailed Kantar product subgroups. Table A.4 reports the six subgroups with the lowest accuracy among those containing at least 300 validation observations.

Table A.4: Product Subgroups with the Lowest Exact-Match Accuracy

Kantar subgroup	Observations	Accuracy
Pastry dough for filling	359	0.267
Other seasonings	313	0.329
Water	1,950	0.340
Pastries	5,934	0.777
Fresh prepared meals	7,631	0.790
Frozen prepared meals	3,931	0.800

Notes: Results pool the 2016–2019 validation samples. Accuracy is the proportion of Kantar products assigned to exactly the correct Agribalyse standardized product. The table is restricted to subgroups containing at least 300 validation observations.

Performance is generally stronger for the high-emission product subgroups for which classification errors would be most consequential. Accuracy is 97.4% for beef, 96.8% for veal, 96.9% for poultry,

93.4% for fish, 92.4% for pork, and 88.9% for lamb and mutton. The weakest performance is instead concentrated in water, some cooking aids, and prepared foods. Errors involving water are likely to have limited consequences for estimated carbon footprints because its emission intensity is low. More generally, the effect of a classification error depends on the difference between the emission intensity factors of the true and predicted standardized products, rather than on the occurrence of an error alone.

A.3 Aggregation of Food Products into Product Categories

Food products are grouped into 13 categories that reflect differences in both the nutritional and carbon content of food. The 13 categories are defined as follows.

1. *Starchy food*: pasta, bread, semolina, cook-type cereals, potatoes.
2. *Fresh fruits and vegetables*: fresh fruits, fresh vegetables.
3. *Red meat*: beef, mutton, lamb.
4. *White meat*: chicken, turkey, veal, rabbit, pork, poultry.
5. *Other meat*: cold cuts, mix of meat, ham, game meat, frogs.
6. *Fish*: fish, seafood, fish eggs, surimi, mix of fish.
7. *Dairy*: goat cheese, cow cheese, sheep cheese, melted cheese, milk, yogurts.
8. *Snacks*: sugary biscuits, jam, honey, chocolate spread, cereal/granola bars, chocolate, pastries, breakfast drink preparation, breakfast cereals, ice cream, desserts, dry fruits and seeds, salty biscuits, olives, crisps.
9. *Ready meals*: prepared dishes, frozen dishes, canned dishes.
10. *Soft drinks*: sodas, syrups, juices.
11. *Other beverages*: coffee, tea, infusions, water, chicory.
12. *Seasonings*: spices, oil, vinegar, butter, croutons, breadcrumbs, raw pastry, confectionery flavorings, flour, cornflour, prepared crust, coconut milk, cream, lemon juice, fresh herbs, dry herbs, garlic, onion.
13. *Non-fresh fruits and vegetables*: canned fruits, canned vegetables, frozen fruits, frozen vegetables, lyophilized vegetables, beans, dried vegetables, packaged vegetables.

An additional category composed of very heterogeneous products, such as baby foods and food supplements, has been excluded from the analysis. Alcohol is also excluded from this analysis.

B Additional Evidence on Food Carbon Footprints

B.1 Food Carbon Footprints and Other Environmental Impacts

To assess how well food carbon footprints proxy for broader environmental impacts, I compute log-log correlations between GHG Emissions (CO_2e) and the other environmental impact indicators from the Agribalyse 3.0 database. Table B.1 reports the results.

Table B.1: Log-log correlations between emission intensity and environmental indicators

Indicator	**Products used**	**Log-log Pearson**	**Spearman ρ**
Ozone depletion	2,480	0.809	0.860
Ionising radiation	2,480	0.656	0.659
Photochemical ozone formation	2,480	0.887	0.934
Particulate matter	2,480	0.948	0.949
Acidification	2,480	0.947	0.947
Terrestrial eutrophication	2,480	0.938	0.937
Freshwater eutrophication	2,480	0.842	0.889
Marine eutrophication	2,480	0.893	0.912
Land use	2,435	0.773	0.733
Freshwater ecotoxicity	2,480	0.802	0.802
Water resource depletion	2,479	0.541	0.532
Energy resource depletion	2,480	0.830	0.859
Mineral resource depletion	2,480	0.780	0.819

Notes: Data from the Agribalyse 3.0 database. The table reports correlations between $\ln(CO_2e)$ and the natural log of each environmental indicator. Emission intensity is measured in kg CO_2 eq/kg product. Rows with missing, zero, or negative values are excluded from each indicator-specific log-log correlation.

B.2 Variation in Emission Intensities and Household Food Carbon Footprints

Table B.2 decomposes the variance of emission intensity into within- and between-group components, and Figure B.1 plots Lorenz curves for total and red-meat emissions.

Table B.2: Decomposition of the Variance of Emission Intensity.

Share of variance for different aggregation levels	Within	Between
All product categories ($N = 13$)	43.2	56.8
Meat categories only ($N = 3$)	28.4	71.6

Notes: This table decomposes the variance of emission intensity into between-group and within-group components, for two levels of product aggregation.

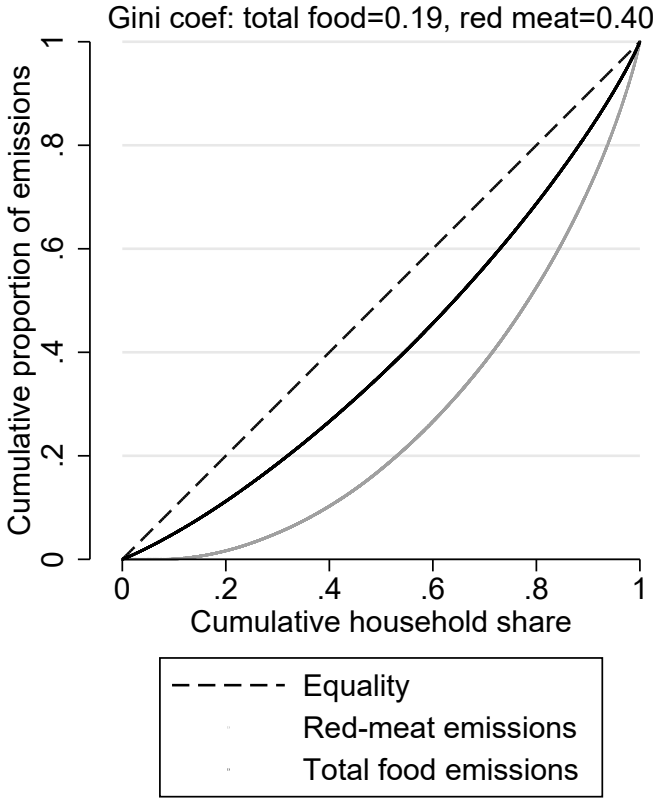


Figure B.1: Lorenz Curves: Total Emissions and Red Meat Emissions.

Notes: Distribution of food related emissions. *Source*: Own calculations based on Kantar WorldPanel 2016-2019 and Agribalyse data.

B.3 Income and Household Characteristics

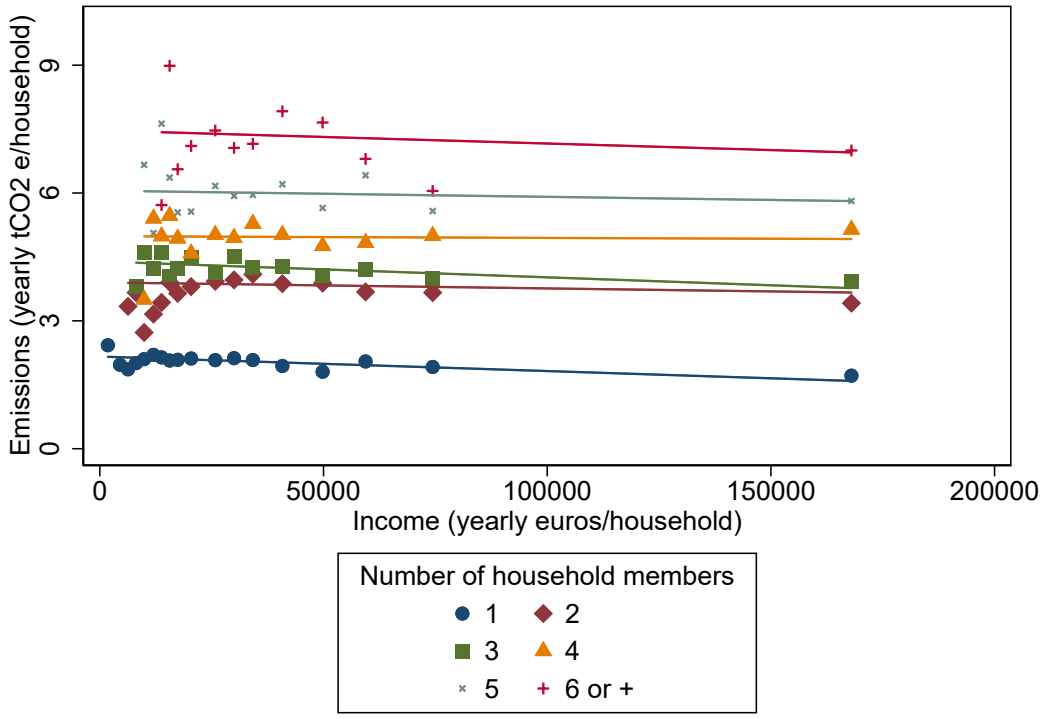


Figure B.2: Income and Food Carbon Footprints for Different household Sizes.

Notes: Binscatter of income and food emissions (20 equally sized bins) per household size. The pattern holds when excluding the last income bin. *Source*: Own calculations based on Kantar WorldPanel 2016-2019 and Agribalyse data.

Table B.3: Household Characteristics and Food Carbon Footprints.

	Log food carbon footprint
	(1)
Education (reference: primary or less)	
Secondary (0/1)	-0.0007
	(0.0191)
Bachelor (0/1)	-0.1109***
	(0.0210)
Master or above (0/1)	-0.1659***
	(0.0215)
Log of Income	0.0098***
	(0.0036)
Age of household head (in years)	0.0073***
	(0.0003)
Household size	0.0749***
	(0.0033)
Rural (0/1)	0.0518***
	(0.0100)
Car owner (0/1)	0.0845***
	(0.0136)
Home owner (0/1)	0.0268***
	(0.0085)
Constant	0.1292***
	(0.0301)
Year fixed effects	Yes
Observations	23,383

Notes: The dependent variable is the log annual household food carbon footprint per consumption unit. Income is measured per consumption unit. Rural, car owner, and home owner are indicator variables. The regression includes year fixed effects and is estimated using annual survey weights. Standard errors are reported in parentheses. *p<0.10, **p<0.05, ***p<0.01.

Table B.4: Food-Basket Composition by Income Decile

	Expenditure share (%)			Volume share (%)			Emissions share (%)			Unit value (€/kg)		
Food category	D1	D5	D10	D1	D5	D10	D1	D5	D10	D1	D5	D10
Red meat	5.51	5.62	5.65	1.48	1.33	1.21	19.41	19.11	19.00	9.96	11.72	15.49
Other meat	9.95	10.00	8.94	2.87	2.60	2.19	9.61	9.60	8.87	9.26	10.64	13.55
White meat	6.94	7.43	7.00	2.62	2.48	2.19	7.34	7.82	7.84	7.08	8.30	10.63
Fish	5.20	6.59	8.05	1.40	1.63	1.81	3.11	4.02	5.05	9.91	11.20	14.72
Ready meals	8.53	6.85	7.86	4.19	3.18	3.50	7.44	5.67	6.32	5.44	5.97	7.45
Snacks	20.43	19.43	17.10	12.96	12.05	10.81	18.45	17.97	16.49	4.22	4.46	5.25
Seasonings	7.35	7.25	6.10	6.32	5.66	4.52	7.23	7.33	6.34	3.11	3.55	4.48
Dairy	10.55	10.11	9.36	10.92	9.55	7.65	9.03	8.80	8.14	2.59	2.93	4.06
Other drinks	4.34	4.68	4.66	18.97	20.75	20.67	5.87	6.24	6.16	0.61	0.62	0.75
Non-fresh fruit and vegetables	3.13	3.07	2.73	4.34	3.80	3.15	2.67	2.28	1.89	1.93	2.24	2.88
Starches	3.78	3.56	3.28	5.53	4.76	4.10	2.60	2.38	2.31	1.83	2.07	2.65
Fresh fruit and vegetables	9.43	11.43	15.75	16.73	23.40	30.71	4.74	6.69	9.66	1.51	1.35	1.70
Soft drinks	4.86	3.98	3.51	11.67	8.82	7.49	2.50	2.08	1.93	1.12	1.25	1.56
All meat	22.40	23.04	21.59	6.98	6.40	5.59	36.36	36.53	35.70			
All animal products	38.15	39.75	39.00	19.30	17.58	15.05	48.50	49.35	48.89			

Notes: D1, D5, and D10 denote the first, fifth, and tenth deciles of household income per consumption unit, respectively. Expenditure, volume, and emissions columns report the share of the corresponding total within each income decile attributable to each food category. Unit values are calculated as expenditure divided by purchased volume, are expressed in deflated euros per kilogram. All meat comprises red, white, and other meat. All animal products additionally include fish and dairy. The two aggregate rows are not included in the column totals. Shares may not sum exactly to 100 because of rounding.

B.3.1 Evidence from Individual Food Consumption: INCA3

I use individual food consumption data from the INCA3 survey to assess whether the weak relationship between income and food carbon footprints also holds in an alternative dataset. This survey, conducted by the French Dietary Agency (ANSES) between 2014 and 2015, covers individual food consumption for a representative sample of the French adult population (N = 2,121).[31] The data report daily volumes of food consumed on a representative day, together with socioeconomic variables. Relative to the Kantar WorldPanel, this dataset has three advantages: (i) it measures consumption rather than purchases, so waste does not inflate measured intake; (ii) it is individual-level rather than household-level; and (iii) it covers total food consumption, both at home and out-of-home. It is, however, a one-time cross-section based on a food diary, and expenditures are not reported, making the Kantar WorldPanel better suited to the analysis of food choices in response to changing prices.

Computing carbon footprints requires matching food consumption in INCA3 to environmental information in Agribalyse. The two datasets contain 2,886 and 2,481 unique labels respectively, each referring to a product cluster. Since they have not previously been matched at the product level, I rely on a mixed procedure combining string matching, hand matching and natural language processing, described in Berland and Leroutier (2025).

Figure B.3 plots mean daily food emissions per individual across ten income categories. Average daily emissions rise from about 8.7 to 9.2 kg CO_2eq between the lowest and highest income category, an increase of roughly 6%, even though monthly income differs by nearly a factor of seven between these two income groups. This confirms the pattern found with the Kantar WorldPanel data, and indicates that it is not driven by the measurement of purchases rather than consumption, by aggregation at the household level, or by the omission of out-of-home food.

[31] The survey initially included 3,157 adults, but only 2,121 participated in face-to-face interviews and provided detailed documentation of their food consumption.

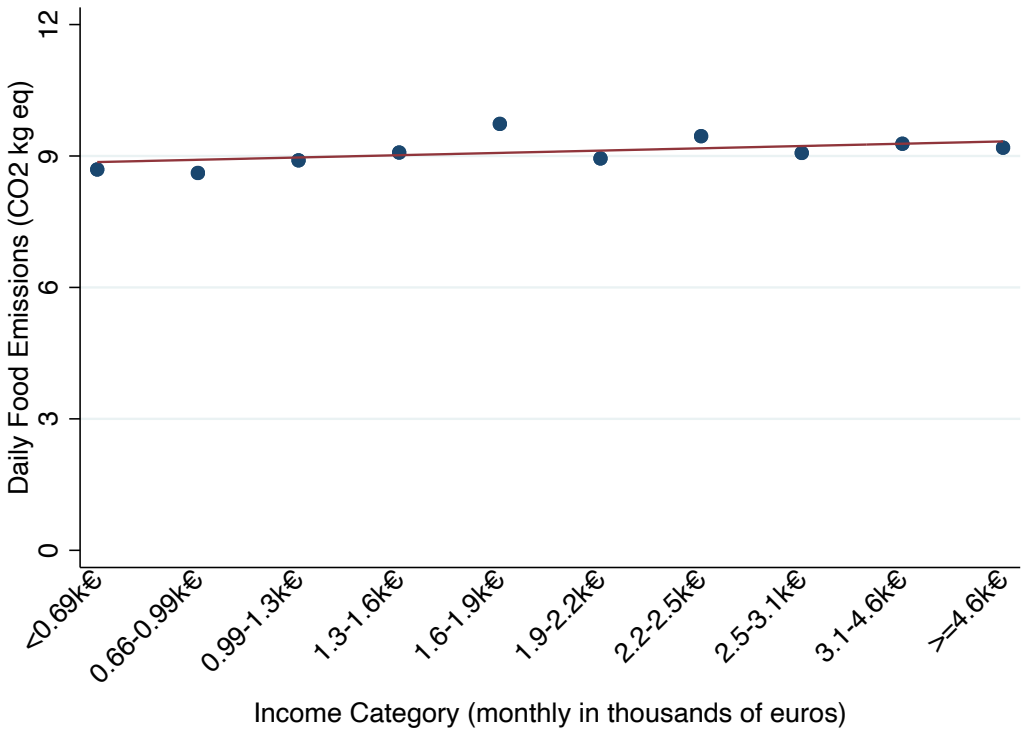


Figure B.3: Income and Food (at-home and out-of-home) Carbon Footprints.

Notes: Sample of 2,121 respondents (individuals) observed on a representative day, sample weights are used. *Source*: INCA3 data (2017) and Agribalyse 2017.

B.4 Quantity-Based and Expenditure-Based Measures

Figure B.4 compares average food carbon footprints by income decile under a quantity-based measure and an expenditure-based measure built from Exiobase3 emission intensities per euro.

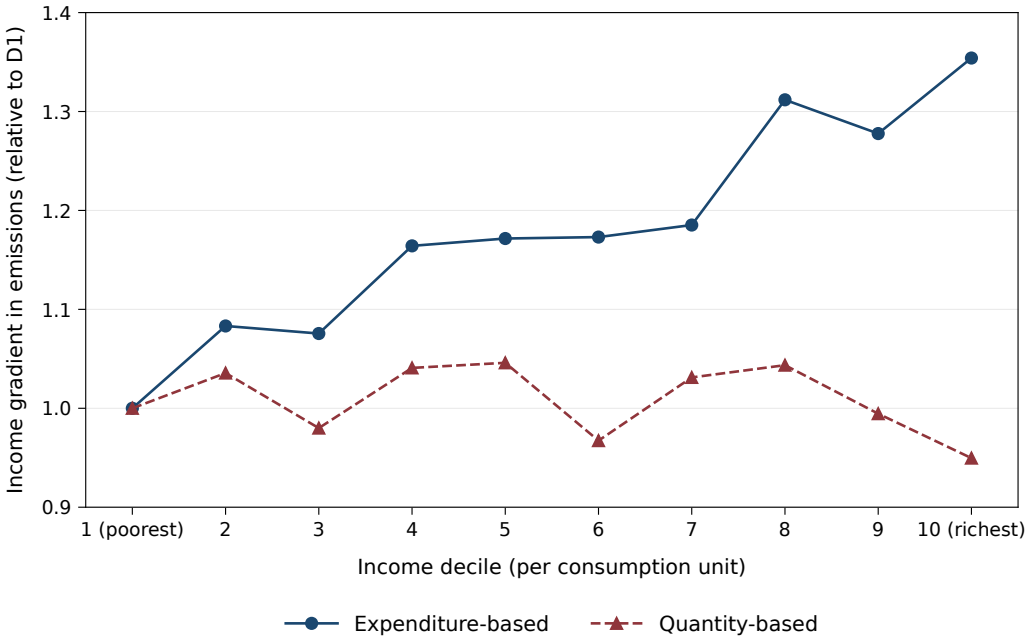


Figure B.4: Bias from using Expenditure based approach to construct Food Carbon Footprints.

Notes: The figure reports average food carbon footprints by household income decile, with income measured per consumption unit. The quantity-based measure combines observed quantities with product-level carbon intensities. The expenditure-based measure combines household expenditure with the corresponding emission intensities per euro from Exiobase3, using the French values for 14 food groups. Of the 25 food and agricultural groups in Exiobase3, live animals and primary agricultural and fishing sectors enter the processed food groups as intermediate inputs and are not bought directly by households, while a few primary groups are retained because they correspond to products bought unprocessed; this leaves 18 groups, of which 14 have a counterpart among the food categories in the data: wheat; cereal grains nec; vegetables, fruit and nuts; oil seeds; raw milk; products of meat cattle; products of meat pigs; products of meat poultry; meat products nec; vegetable oils and fats; dairy products; sugar; fish products; and food products nec (nec denotes "not elsewhere classified").
Every food category in the data is assigned an intensity. Emission intensities per euro are considerably coarser than the standardized products used in the quantity-based measure: the classification names processing sectors defined by their primary input, but has no group for composed foods, so snacks, desserts, ready meals, seasonings and soft drinks all fall into food products nec, which absorbs 40% of the products mapped. Within that group, measured emissions vary only with the prices households pay.
Each series is normalized to one in the lowest income decile. Thus, a value of 1.35 indicates a food carbon footprint 35% larger than that of households in the lowest decile.

B.5 Stated Environmental Preferences and Food Emissions

Table B.5 summarizes stated environmental preferences from the 2019 Kantar WorldPanel attitudinal survey, and Figure B.5 relates them to household food carbon footprints.

Table B.5: Summary Statistics for Stated Environmental Preferences, 2019

Statement	N	Mean	Std. dev.	Min.	Max.
Eat less meat	4,744	2.74	0.87	1	4
Buy fewer products to avoid waste	4,744	3.06	0.68	1	4
Preserve the planet's resources	4,744	3.09	0.66	1	4
Buy eco-friendly products	4,744	3.02	0.69	1	4

Notes: Data come from the 2019 Kantar WorldPanel attitudinal survey matched to food purchases at the household level. The household head reports agreement with each statement on a four-point Likert scale, from 1 (fully disagree) to 4 (fully agree). Means and standard deviations use annual household sampling weights.

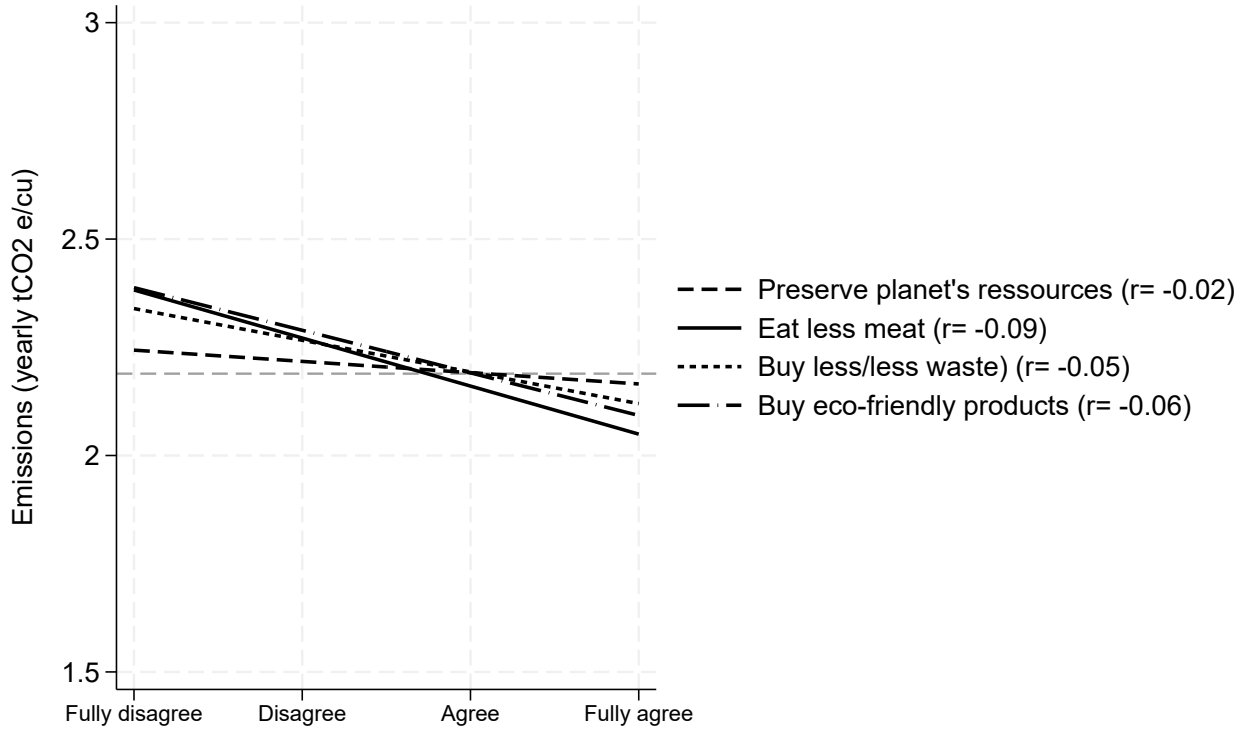


Figure B.5: Stated Environmental Preferences and Food Carbon Footprints, 2019.

Notes: Data come from the 2019 Kantar WorldPanel attitudinal survey matched to household food purchases. The analysis includes 4,744 households. Lines show linear fits between agreement with each statement and annual food emissions per consumption unit. Responses range from 1 (fully disagree) to 4 (fully agree). In the legend, r denotes the correlation coefficient between households responses and annual food carbon footprints per consumption unit. Annual household sampling weights are used.

C Demand Estimation

C.1 Demand Elasticities

I recover Marshallian and Hicksian price elasticities, as well as expenditure elasticities, from the estimated QUAIDS parameters, following Lecocq and Robin (2015). The elasticities are evaluated at sample means.

Let $z_{it} = \ln X_{it} - a(\mathbf{p}_{it}; \theta)$ denote log real food expenditure. Differentiating the budget-share equation gives

$$\mu_{ijt} = \beta_j + 2\lambda_j \frac{z_{it}}{b(\mathbf{p}_{it}; \theta)}, \tag{C.1}$$

$$\mu_{ijkt} = \gamma_{jk} - \mu_{ijt}\left(\alpha_k + \sum_{\ell=1}^{J} \gamma_{k\ell} p_{i\ell t}\right) - \lambda_j \beta_k \frac{z_{it}^2}{b(\mathbf{p}_{it}; \theta)}. \tag{C.2}$$

The expenditure elasticity is

$$\eta_{ijt} = 1 + \frac{\mu_{ijt}}{s_{ijt}}. \tag{C.3}$$

The Marshallian (uncompensated) price elasticity is

$$\varepsilon^M_{ijkt} = \frac{\mu_{ijkt}}{s_{ijt}} - \delta_{jk}, \tag{C.4}$$

where δ_{jk} equals one if $j = k$ and zero otherwise.

Hicksian (compensated) price elasticities follow from the Slutsky equation:

$$\varepsilon^H_{ijkt} = \varepsilon^M_{ijkt} + s_{ikt}\eta_{ijt}. \tag{C.5}$$

C.2 Comparison with Estimates from the Literature

Table C.1 reports expenditure and price elasticities from product-space demand models estimated in the literature.

Table C.1: Estimates from the Literature (Product Space Models).

	Amano-Patino (2019)			Caillavet et al. (2016)		
	$\hat{\eta}_E$	$\hat{\eta}_{CP}$	$\hat{\eta}_{UP}$	$\hat{\eta}_E$	$\hat{\eta}_{CP}$	$\hat{\eta}_{UP}$
Processed meat	0.5413	-1.2145	-1.2161			
Fresh meat and seafood	0.3741	-1.1864	-1.1874			
Fruit and Vegetables	0.6012	-0.9787	-0.9810			
Dairy	0.3572	-0.6985	-0.7012	0.952	-1.347	-
Beef				1.002	-1.113	-
Country	US			France		

Notes: $\hat{\eta}_E$ is the estimated expenditure elasticity, and $\hat{\eta}_{UP}$ (resp. $\hat{\eta}_{CP}$) the estimated uncompensated (resp. compensated) price elasticity.

C.3 Matrix of Own- and Cross-Price Elasticities

Table C.2 reports the full matrix of compensated own- and cross-price elasticities.

Table C.2: Compensated own- and cross-price elasticities

Demand category	Starches	Fresh fruit/veg.	Red meat	White meat	Other meat	Fish	Dairy	Snacks	Ready meals	Soft drinks	Seasonings	Other drinks	Non-fresh fruit/veg.
Starches	-1.216***	0.103***	0.298***	-0.040	0.077	0.079*	0.261***	0.189**	0.046	0.129***	0.022	-0.050	0.015
	(0.053)	(0.020)	(0.047)	(0.055)	(0.056)	(0.045)	(0.085)	(0.082)	(0.054)	(0.050)	(0.031)	(0.032)	(0.016)
Fresh fruit/veg.	-0.010	-0.279***	-0.261***	-0.349***	-0.218**	0.356***	0.077	-0.095	0.008	-0.111	-0.137**	0.082	-0.037
	(0.102)	(0.084)	(0.092)	(0.111)	(0.110)	(0.092)	(0.163)	(0.159)	(0.108)	(0.099)	(0.061)	(0.064)	(0.034)
Red meat	0.138**	0.119***	-1.410***	0.054	-0.018	0.032	0.056	0.209**	0.010	0.009	-0.016	0.052	0.028
	(0.064)	(0.025)	(0.058)	(0.066)	(0.068)	(0.054)	(0.102)	(0.099)	(0.066)	(0.060)	(0.037)	(0.039)	(0.020)
White meat	-0.037	0.071***	0.062	-1.073***	0.080	0.103**	0.085	-0.004	-0.014	0.097*	0.004	0.013	-0.012
	(0.054)	(0.021)	(0.047)	(0.056)	(0.057)	(0.046)	(0.086)	(0.084)	(0.056)	(0.051)	(0.032)	(0.033)	(0.017)
Other meat	0.035	-0.018	0.093**	0.179***	-0.837***	-0.012	0.089	0.179***	0.127***	0.021	0.095***	0.053**	0.026*
	(0.045)	(0.019)	(0.039)	(0.046)	(0.047)	(0.038)	(0.071)	(0.069)	(0.046)	(0.042)	(0.026)	(0.027)	(0.014)
Fish	0.023	0.555***	-0.205**	-0.106	-0.146	-0.923***	0.114	-0.080	0.118	-0.107	-0.023	0.160***	0.011
	(0.093)	(0.057)	(0.083)	(0.098)	(0.100)	(0.083)	(0.149)	(0.145)	(0.098)	(0.088)	(0.056)	(0.058)	(0.030)
Dairy	0.131***	-0.057***	0.188***	0.258***	0.224***	-0.003	-1.107***	0.399***	0.221***	0.079**	0.132***	0.080***	0.065***
	(0.043)	(0.022)	(0.039)	(0.045)	(0.046)	(0.037)	(0.068)	(0.067)	(0.045)	(0.040)	(0.025)	(0.026)	(0.014)
Snacks	0.058*	-0.035*	0.210***	0.160***	0.170***	-0.027	0.141**	-0.634***	0.068*	0.042	0.143***	0.019	0.049***
	(0.034)	(0.018)	(0.031)	(0.036)	(0.037)	(0.030)	(0.055)	(0.053)	(0.036)	(0.033)	(0.021)	(0.021)	(0.011)
Ready meals	0.046	-0.011	0.143**	0.142*	0.262***	0.061	0.247**	0.201*	-1.087***	0.096	0.165***	0.092**	0.039
	(0.075)	(0.032)	(0.066)	(0.077)	(0.081)	(0.064)	(0.120)	(0.116)	(0.077)	(0.071)	(0.044)	(0.046)	(0.024)
Soft drinks	0.111*	-0.040	0.124**	0.300***	0.050	-0.054	0.044	0.038	0.094	-0.823***	0.134***	0.036	0.028
	(0.066)	(0.027)	(0.058)	(0.068)	(0.071)	(0.056)	(0.106)	(0.102)	(0.068)	(0.063)	(0.039)	(0.041)	(0.021)
Seasonings	0.043	-0.141***	0.186***	0.240***	0.239***	-0.025	0.086	0.369***	0.149***	0.131***	-0.874***	0.056**	0.075***
	(0.041)	(0.024)	(0.037)	(0.043)	(0.044)	(0.036)	(0.065)	(0.065)	(0.044)	(0.039)	(0.024)	(0.026)	(0.013)
Other drinks	-0.037	0.102***	0.112*	0.063	0.172**	0.160***	0.202*	0.139	0.180**	0.057	0.131***	-1.183***	0.070***
	(0.072)	(0.028)	(0.063)	(0.074)	(0.077)	(0.061)	(0.115)	(0.111)	(0.074)	(0.068)	(0.042)	(0.044)	(0.022)
Non-fresh fruit/veg.	0.024	0.032	0.153***	0.074	0.087	0.061	0.093	0.181*	0.040	0.040	0.109***	0.076**	-0.925***
	(0.061)	(0.025)	(0.054)	(0.063)	(0.065)	(0.052)	(0.098)	(0.095)	(0.063)	(0.058)	(0.036)	(0.038)	(0.019)

Notes: Entries are compensated price elasticities evaluated at the full-sample average. Rows indicate the demand category and columns indicate the category whose price changes. Significant positive off-diagonal elasticities indicate net substitutes, and significant negative ones net complements. For instance, when the price of fresh fruit and vegetables increases by 1% (column 2), the demand for starches increases by 0.103% (column 2, row 1). Conversely, when the price of starches increases by 1% (column 1), the demand for fresh fruit and vegetables does not change significantly (-0.010%, column 1, row 2). Standard errors calculated with the delta method are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C.4 Robustness to Demand-System Restrictions

As a sanity check, I estimate a budget-share regression using 2SLS, with the same 13 category-level leave-one-out price indices, household characteristics, and département and quarter fixed effects as in the AIDS model, where log food expenditure is instrumented by household income. For red meat, I estimate:

$$w_{i3t} = \alpha_3 + \sum_{j=1}^{13} \theta_{3j} \ln P_{ijt}^{\mathrm{LOO}} + \beta_3 \ln X_{it} + \mathbf{Z}_{it}'\boldsymbol{\delta}_3 + \mu_{d(i)} + \lambda_t + \varepsilon_{i3t}, \tag{C.6}$$

where w_{i3t} is household i's red-meat budget share in quarter t, P_{ijt}^{LOO} is the leave-one-out price index of category j defined in equation (4), and X_{it} is total food expenditure, instrumented using household income. The vector $\mathbf{Z}_{it}$ includes household size, rural residence, car ownership, and homeownership. The terms $\mu_{d(i)}$ and λ_t denote département and quarter fixed effects, respectively. Both specifications use the same expenditure weights.

Appendix Table C.3 presents the estimates from the AIDS and the budget share regressions. Unlike the AIDS model, the single-equation regression does not impose symmetry, homogeneity, or adding-up restrictions. The estimated own-price coefficient for red meat is $\hat{\theta}_{33} = -0.034$, close to the corresponding AIDS parameter $\hat{\gamma}_{33} = -0.029$.

For most other categories, the estimates are also broadly comparable, particularly for starches, fish, dairy, ready meals, and other drinks. Larger differences arise for a few categories, although the corresponding single-equation estimates are generally small and not statistically significant. Overall, the results suggest that the estimated price coefficients are not very sensitive to the restrictions imposed by the demand system.

Table C.3: Own-price coefficients across food categories: AIDS and separate IV regressions

Food category	AIDS γ_{ii}	Budget share coefficient
Starches	-0.0092	-0.0095***
		(0.0036)
Fresh fruit/veg.	0.0112	0.0001
		(0.0025)
Red meat	-0.0291	-0.0341***
		(0.0065)
White meat	-0.0069	-0.0028
		(0.0073)
Other meat	0.0037	-0.0008
		(0.0059)
Fish	-0.0073	-0.0058
		(0.0049)
Dairy	-0.0269	-0.0307***
		(0.0099)
Snacks	0.0127	0.0049
		(0.0141)
Ready meals	-0.0119	-0.0157**
		(0.0079)
Soft drinks	-0.0061	0.0025
		(0.0046)
Seasonings	0.0027	-0.0016
		(0.0022)
Other drinks	-0.0108	-0.0109***
		(0.0027)
Non-fresh fruit/veg.	0.0012	0.0013
		(0.0008)

Notes: The first column reports the diagonal elements of the estimated AIDS gamma matrix. The second reports coefficients from separate 2SLS regressions of each category's expenditure share on all category price indices, instrumenting log food expenditure with log income per consumption unit. Regressions include household characteristics, quarter and département fixed effects, and annual analytic weights. Standard errors in parentheses are clustered at the household level and are reported for the IV regressions only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.